BOUNDARY GLOTTALS AND A'INGAE INFORMATION STRUCTURE: A MORPHOLOGICAL ARGUMENT FOR A DISCOURSE FEATURE GEOMETRY

MAKSYMILIAN DĄBKOWSKI
UNIVERSITY OF CALIFORNIA, BERKELEY
MANUSCRIPT *as of* NOVEMBER 7, 2025

ABSTRACT I describe and analyze a pattern of morphosyntactically conditioned realization of a glottal stop morpheme-*ʔ* in A'ingae (or Cofán, an endangered Amazonian isolate, ISO 639-3: con). The glottal stop -*ʔ* appears before three information structural morphemes—the new topic -*ta* NEW, contrastive topic -*ja* CNTR, and exclusive focus -*yi* EXCL—when they attach directly to a TP, but not when they attach to other categories, such as DPs, CPs, or adverbs. I propose that the glottal stop (-*ʔ*) is a spell-out of T° conditioned by linear adjacency to a higher-order *discourse* feature [δ] (Bossi and Diercks, 2019; Mikkelsen, 2015) that dominates all the maximal information structural features in a feature geometry. By documenting an overt realization of a vocabulary item conditioned by [δ], the analysis provides new morphological evidence for a hierarchical arrangement of discourse features (Bossi and Diercks, 2019; Mikkelsen, 2015), and of Ā-feature geometry more broadly (e. g. Aravind, 2018; Baclawski, 2019; Baier, 2018). Additionally, by reducing the cross-linguistically unique distribution of the A'ingae -*ʔ* to independently motivated theoretical notions (e. g. Embick, 2010, 2015; Shlonsky, 1997), it bears out a novel prediction of syntax-centric models of the morphological grammar.

1 INTRODUCTION

This paper describes and analyzes a pattern of morphosyntactically conditioned realization of the glottal stop -*ʔ* in A'ingae (or Cofán, an endangered Amazonian isolate, ISO 639-3: con). A'ingae has four information structure (IS) markers: the new topic -*ta* NEW,[1] contrastive topic -*ja* CNTR, exclusive focus -*yi* EXCL, and additive focus -*ʔkhe* ADD. The first three of these

1 The following glossing abbreviations have been used: 1 = first person, 2 = second person, 3 = third person, ACC = accusative, ACC2 = accusative 2, ADD = additive focus, ADJ = adjectivizer, ADN = adnominal, ADV = adverbializer, AIMP = attenuated imperative, ANA = anaphora, AND = andative, ANIM = animate, APPR = apprehensional, ASSR = assertive, ATTR = attributive, AUX = auxiliary, BND = bounded, C = complementizer, CAUS = causative, CNTR = contrastive topic, DAT = dative, DEF = definite, DEM = demonstrative, DIST = distal, DRN = diurnal, DS = different subject, ELAT = elative, EN = event nominalizer, EXCL = exclusive focus, FLAT = flat, FOC = focus, FRST = frustrative, HN = habitual nominalizer, HORT = hortative, IF = conditional, IMP = imperative, INF = infinitive, INGR = ingressive, INS = instrumental, IPFV = imperfective, IRR = irrealis, ITER = iterative, LAT = lateral, LOC = locative, NEG = negative, NEW = new topic, NIN = negative individual nominalizer, OBJ = object, OP = operator, PASS = passive, PERM = permissive, PL = plural, PLA = pluractional, PLC = place, PLS = plural subject, PROH = prohibitive, PRSP = prospective, PST = past, PURP = purposive, RCPR = reciprocal, REL = relative, RPRT = reportative, SG = singular, SML = similative, SN = subject nominalizer, SS = same subject, STA =

markers show a regular alternation between plain realizations (i. e. without a preceding glottal stop; *-ta*, *-ja*, *-yi*) and preglottalized realizations (*-ʔta*, *-ʔja*, *-ʔyi*), depending on the syntactic category of the base of attachment.

When the IS morphemes are attached to phrases of most syntactic categories, including—for example—noun phrases (1a), overtly subordinated clauses (1b), and adverbs (1c), no glottal stop is realized. However, when they are attached to infinitive verbs (2a), finite predicates in subordinate clauses (2b), or negative adverbial clauses (2c), in subordinate clauses, they are realized as preglottalized. The key pattern is illustrated below with the new topic *-ta* NEW.[2] Anticipating my analysis, I gloss the glottal stop as a separate morpheme *-ʔ* T. The relevant IS markers (and the preceding glottal stop, when present) are underlined throughout the paper.[3]

(1) NEW TOPIC *-TA* (*-NDA*) REALIZED AS PLAIN

a. ATTACHED TO A NOUN PHRASE

yáya=ta =tsû tsámpi=ni já
dad=NEW =3 forest=LOC go
"Dad went hunting." (2024-06-10(1)_mll)

b. ATTACHED TO AN OVERTLY SUBORDINATED VERB

kháʔna-saʔne-nda =ngi íyûʔû
steal-APPR-NEW =1 scold
"I advised (them) not to steal." (2024-05-31(1)_sia)

c. ATTACHED TO AN ADVERB

káni-nda =tsû án
yesterday-NEW =3 eat
"S/he ate yesterday." (2025-07-22(1)_mll)

(2) NEW TOPIC *-ʔTA* (*-ʔNDA*) REALIZED WITH PREGLOTTALIZATION

a. ATTACHED TO AN INFINITIVE VERB

án-ñe-ʔ-nda =ngi séʔpi
eat-INF-T-NEW =1 forbid
"I forbade eating." (2024-05-31(1)_sia)

spatial/temporal anaphora, T = tense, TENT = tentative, THUS = manner demonstrative, TOP = topic, VDM = verbal diminutive, VEN = venitive, VPA = VP anaphor, WH = content interrogative, YNQ = polar interrogative.

2 A'ingae shows iterative progressive nasal spreading (as well as non-iterative regressive nasalization) (Bennett et al., 2025; Dąbkowski, 2024a; Sanker and AnderBois, 2024). Consequently, *-ta* NEW, *-ja* CNTR, and *-yi* EXCL surface as *-nda* NEW, *-jan* CNTR, and *-ñi* EXCL after nasal vowels. This oral-nasal alternation is orthogonal to the morphosyntactically conditioned preglottalization under scrutiny.

3 Many functional morphemes, including the four IS markers under investigation, can attach to constituents of various categories, including noun phrases—in which case they appear at their very right edge, regardless of phrase-internal word order—as well as verb phrases and adverbs—in which case they always appear on the head of the phrase, i. e. the verb or adverb itself. Correspondingly, I represent them as either clitics or affixes. This distinction does not, in itself, have any morphophonological correlates (Dąbkowski, 2021b, 2024b), nor does it correlate directly with the realization or non-realization of *-ʔ* T.

b. ATTACHED TO A FINITE PREDICATE
sése *jí-ʔ-ta* *=tsû* *án-ña*
papagayo come-T-NEW =3 eat-IRR
"If the papagayo comes, it will eat." (2024-06-10(1)_mll)

c. ATTACHED TO A NEGATIVE ADVERBIAL CLAUSE
setsa-yé-mbe-ʔ-ta *=tsû* *kánse*
infect-PASS-ADV.NEG-T-NEW =3 live
"S/he does not get sick." (2025-07-17(1)_mll)

The four markers under discussion (*-ta* NEW, *-ja* CNTR, *-yi* EXCL, *-ʔkhe* ADD) form a natural class—they all encode information structural meanings, attach to the same range of constituents, and always appear at the very end of a phrase, occupying the same morphosyntactic position. (For more on their similarities, see Section 3.) As such, the realization of *-ʔ* T should be seen not as conditioned by a set of unrelated morphemes, but rather attributed to some morphosyntactic property that all of them have in common.

In Section 6, I will propose that the glottal stop *-ʔ* is a realization of the T-head conditioned by linear adjacency to IS morphology. The conditioning environment is formalized as a higher-order *discourse* feature [δ] (Bossi and Diercks, 2019; Mikkelsen, 2015), which is entailed by all the more specific IS features, including new topic [NEW], contrastive topic [CNTR], exclusive focus [EXCL], and additive focus [ADD]. The proposed feature geometry is visualized in (3).

(3) DISCOURSE FEATURE GEOMETRY

δ

NEW CNTR EXCL ADD

Since all the information structural markers inherit from [δ], they all satisfy the T-head's conditioning environment. However, if any phonologically overt material appears between the T-head and the IS marker, the realization of the glottal stop *-ʔ* is blocked (Embick, 2010, 2015). A structural schema representing the conditions for the realization of *-ʔ* is given in (4). (Conversely, in all the cases where *-ʔ* does not appear, these conditions are not met—i. e. there is no T-head directly adjacent to an IS marker.)

(4) ʔ-REALIZATION SCHEMA

right edge of TP ↘
[*root* … ***-ʔ*** $]_{\mathrm{TP}}$ *-sfx*$_{[\delta]}$
↑ ↑ ↖ information structural morpheme
T° no overt material intervening

The implications of this analysis are twofold. First, in previous literature, the notion of *discourse differentiation* (Mikkelsen, 2015), formalized by Bossi and Diercks (2019) as a superordinate discourse feature [δ], has been motivated by word-order facts in Kipsigis (Nilo-Saharan, Kenya; Bossi and Diercks, 2019) and Danish (North Germanic, Denmark; Mikkelsen, 2015). The present study focuses on patterns of allomorphy in the domain of A'ingae information structural markers. By showing that all the A'ingae IS morphemes pattern alike in triggering an allomorphy of the T-head, the current paper provides novel morphological evidence for a hierarchical arrangement of discourse features (with [δ] inherited by the maximal topic and focus features), and of Ā-feature geometry more broadly (e. g. Aravind, 2018; Baclawski, 2019; Baier, 2018).

Second, the distribution of *-ʔ* T is typologically unusual—the suffix realizes an abstract morphosyntactic position when adjacent to a marker drawn from a semantically defined set. To the best of my knowledge, no other language has been described as having morphological exponents with comparable characteristics. Nevertheless, the analysis in Section 6 will rely only on (a combination of) previously motivated solutions (e. g. Bossi and Diercks, 2019; Mikkelsen, 2015; Shlonsky, 1997), without introducing any new theoretical devices. This reduction of a cross-linguistically unique pattern to independently needed theoretical atoms shows that the distribution of the A'ingae *-ʔ* T can not only be accounted for, but is in fact predicted by syntax-centric models of word-internal morphological structure such as Distributed Morphology (DM) (e. g. Embick, 2010, 2015; Embick and Noyer, 2007; Halle and Marantz, 1993, 1994).

The rest of the paper is structured as follows. Section 2 provides background on the language, summarizes relevant aspects of its grammar, and lists previous descriptive and theoretical literature. Section 3 introduces the A'ingae four discourse markers, discusses their uses, and presents various morphological and syntactic restrictions on their distribution. Section 4 lays out the structure of A'ingae predicates. Section 5 describes the patterns of *ʔ*-realization observed before the IS markers, and motivates some analytical choices made along the way. Section 6 presents an account of the patterns formalized in DM. Section 7 discusses the findings and concludes.

2 LANGUAGE BACKGROUND

A'ingae (or Cofán, ISO 639-3: `con`) is an endangered Amazonian language spoken by ca. 1,500 Cofán people in the northeast Ecuadorian province of Sucumbíos and the southern Colombian department of Putumayo. The language remains classified as an isolate (AnderBois, Emlen, et al., 2019), despite some previous (mostly geography-driven) claims to the contrary (e. g. in Rivet, 1924, 1952; Ruhlen, 1987).

A'ingae syllable structure can be schematized as (C)V(V)(ʔ)—onsets are optional, nuclei are maximally diphthongal, and glottal stops are the only licit coda. Additionally, glottal stops are disallowed word-finally. The present paper uses the practical orthography with two deviations: glottal stops are represented with the International Phonetic Alphabet (IPA) symbol (*ʔ*), not apostrophe (´), and stress is marked with the acute accent (´). For more on the A'ingae orthography, see Dąbkowski (2024a), Fischer and Hengeveld (2023), and Repetti-Ludlow et al. (2019).

A'ingae is a highly agglutinating, exclusively suffixing, and encliticizing language. The vast majority of the language's functional morphemes are -CV or -ʔCV monosyllables (the preglottalization, if present, is parsed as the coda of the preceding syllable); most of the language's glottal stop tokens come from preglottalized suffixes. There are many lexically contrastive morphemes that constitute plain–preglottalized minimal pairs, e. g. the flat classifier *-je* FLAT vs. imperfective *-ʔje* IPFV, the accusative *=ma* ACC vs. frustrative *-ʔma* FRST, or the dative *=nga* DAT vs. andative *-ʔnga* AND. The glottal stop *-ʔ* T discussed in this paper is different in that it is a separate morpheme. (Notwithstanding, out of terminological convenience, I often refer to the sequences of *-ta*, *-ja*, *-yi* and *-ʔta*, *-ʔja*, *-ʔyi* as—respectively—plain and preglottalized realizations of NEW, CNTR, and EXCL.)

The A'ingae default word order is subject–object–verb (SOV), although in matrix clauses, it is largely free. (Co)subordinate clauses are predominantly verb-final (for some exceptions, see Dąbkowski, 2025c). The language has nominative-accusative morphosyntactic alignment; subjects are unmarked and direct objects are marked with one of two accusative clitics: *=ma* ACC or *=ve* ACC2.[4] A'ingae supports robust pro-drop; any of the clausal arguments may be dropped if inferrable from context.

The data presented in this paper comes from published materials written originally in A'ingae and fieldwork elicitation conducted by the author. Elicitation tasks included translation and grammaticality judgments. All the data drawn from previous publications are cited as such.[5] All the fieldwork data has been deposited in the California Language Archive (CLA) as Dąbkowski (2020) and cited with a `YYYY-MM-DD(N)_ccc`identifier.[6]

Previous descriptive work on the language includes Fischer and Hengeveld (2023) and Hengeveld and Fischer (in prep.) and chapters in Dąbkowski's (2025c) dissertation. Aspects of the language's phonetics and phonology have been analyzed by Dąbkowski (2023c, 2024a, 2025a,b), Repetti-Ludlow et al. (2019), Repetti-Ludlow (2021), and Sanker and AnderBois (2024). Various topics in the language's morphology, syntax, semantics, pragmatics, and their interfaces have been discussed by e. g. AnderBois, Altshuler, and Silva (2023), Dąbkowski and AnderBois (2024, t.a.), Fischer (2007), Fischer and van Lier (2011),

4 The choice of *=ma* ACC or *=ve* ACC2 depends on a variety of factors, including the verb's meaning, its arbitrary selectional restrictions, and properties of the object. For one proposal discussing the semantic differences between *=ma* ACC and *=ve* ACC2, see Karsten (2021).

5 Since glottalization is not reliably represented in written A'ingae (and stress is often not represented at all), I supplied this information by reeliciting the relevant data. At times, my consultants improved on the word choice or corrected some of the grammatical forms. In all such cases, the reported data reflect the choices and judgments of my consultants, not the original medium.

6 `YYYY` stands for the year, `MM` — the month, `DD` — the day, `N` — the number of the consecutive elicitation session conducted on a given day, and `ccc` — the three-letter consultant name abbreviation. File names of the CLA-deposited materials often also contain keywords, briefly summarizing the contents of the session.

Hengeveld and Fischer (2018), Kalpande (2024), Morvillo and AnderBois (2022), and Zheng and AnderBois (2023).

A'ingae glottal stops enter into complex, morphologically mediated interactions with metrical stress, discussed extensively by Dąbkowski (2021b, 2023a, 2024b, 2025c,e, in prep.). The phonological processes involving glottalization take place only in the morphosyntactic projections closest to the root (such as VceP and AspP). Crucially, for the purposes of this paper, glottal stops introduced in later projections (such as the TP and CP relevant here) are not conditioned by phonological factors and have no effect on the phonology of their bases. As such, the factors determining the realization of *-ʔ* T are purely morphosyntactic. Correspondingly, I will not discuss A'ingae glottal morphophonology.

3 A PRIMER ON DISCOURSE MARKERS

A'ingae has four morphemes dedicated to encoding discourse-related information: two encoding topichood, and two—focus. In this section, I introduce the four markers briefly. Further discussion of their uses is given in Section A.1.

The topic of a sentence is the entity that the sentence is intuitively "about." This is to say, the topic is a referent that the rest of the sentence comments on (Krifka, 2008). A'ingae distinguishes two types of overtly marked topics: the new topic *-ta* NEW and the contrastive topic *-ja* CNTR.[7] The new topic marker *-ta* NEW indicates that the referent was not previously present in the discourse. For example, the marker often appears on individuals that are introduced for the first time (5).

(5) REFERENT INTRODUCED WITH *-TA* NEW
[seeing a city in the distance for the first time:]
jú-va *tútu-a=ta* *=ti* *patû́-ndûʔndû*
DIST-DEM white-ADN=NEW =YNQ rock-beach
"Is that white over there a rocky beach?"
(Borman and Chica Umenda, 1982b, p. 24; `2025-08-26(1)_mll`)

The contrastive topic marker *-ja* CNTR is used in the presence of alternative topics in the discourse, especially when information about different individuals is overtly or covertly juxtaposed. For example, it may be used on a subject which contrasts with other discourse participants in whether they took part in some activity (6). In comparable circumstances, English may distinguish the contrasted constituent prosodically.

(6) SUBJECT PROPERTY CONTRASTED WITH *-JA* CNTR
[dialogue between A and B:
A: Who all went?
B: Dad, my uncle, and my older brother.]

7 The semantic contributions of the markers are often not reflected in translations to Spanish or English. Moreover, in most sentences, *-ta* NEW and *-ja* CNTR can be removed without resulting in ungrammaticality.

A: *míngaʔu-si =ki ké=ja já-mbi*
why-DS =2 2SG=CNTR go-NEG
"Why didn't *you* go?"
(Borman and Chica Umenda, 1982b, p. 9; `2025-08-26(1)_mll`)

Focus indicates that in addition to the referent denoted by the marked phrase, other entities are also relevant to the interpretation of the sentence (Krifka, 2008, p. 247). A'ingae has two focus morphemes: the exclusive focus *-yi* EXCL and the additive focus *-ʔkhe* ADD. The exclusive focus *-yi* EXCL indicates that a proposition holds of the marked entity to the exclusion of alternatives; it can often be translated as "only" (7), "just," "very (same)," "as soon as," or with cleft constructions.

(7) EXCLUSIVE FOCUS *-YI* EXCL
tsá-ʔka-en pá-ʔfa-si khuánifae tsandié=ndekhû=yi khûshá-ʔfa
ANA-SML-ADV die-PLS-DS three man=PL.ANIM=EXCL survive-PLS
"When they thus died, only three men were spared."
(Blaser and Chica Umenda, 2008, p. 37; `2024-06-07(1)_mll`)

The additive focus *-ʔkhe* ADD indicates that the proposition holds of the marked entity in addition to some alternatives; it can often be translated as "too," "as well," "also" (8), "and," or "even (though)," and—in negative polarity contexts—as "either" or "(not) even." The additive focus *-ʔkhe* ADD will be analyzed as underlyingly preglottalized, so the overt presence or absence of *-ʔ* T can never be observed before it (see Section 6 for the full analysis). Nevertheless, I present examples with *-ʔkhe* ADD throughout for the sake of completeness. For a more detailed discussion of the uses of *-ta* NEW, *-ja* CNTR, *-yi* EXCL, and *-ʔkhe* ADD, see Section A.1.

(8) ADDITIVE FOCUS *-ʔKHE* ADD
tsá=ma áthe-pa fáesû=ma=ʔkhe áthe
ANA=ACC see-SS other=ACC=ADD see
"When (he) looked, (he) also saw another one."
(Blaser and Chica Umenda, 2008, p. 38; `2024-06-07(1)_mll`)

The two topic markers *-ta* NEW and *-ja* CNTR are mutually incompatible. The exclusive focus *-yi* EXCL may be followed by the additive focus *-ʔkhe* ADD (9a), or by either topic marker (9b-c). Finally, the additive focus *-ʔkhe* ADD may be followed by the new topic *-ta* NEW (9d). The co-occurrence possibilities among the A'ingae IS markers are summarized in Table 1.[8] (These co-occurrence possibilities will not play a role in the analysis, but I am providing them here for completeness of description.)

8 Fischer and Hengeveld (2023) also report that the additive focus marker *-ʔkhe* ADD may be followed by the contrastive topic *-ja* CNTR. Since I have not been able to corroborate this combination, I do not report it here.

(9) Licit combinations of informational structural markers

a. EXCLUSIVE FOCUS *-YI* EXCL + ADDITIVE FOCUS *-ʔKHE* ADD

tsá-ʔma tsé-ki kúse ni fae ávû=ve=yi=ʔkhe índi-ʔfa-mbi =ngi
ANA-FRST STA-DRN night neither one fish=ACC2=EXCL=ADD catch-PLS-NEG =1
"But that night we did not catch even a single fish."
(John 21:3; `2024-06-07(1)_mll`)

b. EXCLUSIVE FOCUS *-YI* EXCL + NEW TOPIC *-TA* NEW

tsá-ʔka-en kánse-ʔfa-ʔ-yi-ta =ki ñuáʔme shaká-ʔfa-ya-mbi
ANA-SML-ADV live-PLS-T-EXCL-NEW =2 truly lack-PLS-IRR-NEG
"Only if you live like this, you will truly not lack (anything)."
(Thessalonians 4:12; `2024-06-07(1)_mll`)

c. EXCLUSIVE FOCUS *-YI* EXCL + CONTRASTIVE TOPIC *-JA* CNTR

tsá-ʔkan-si =tsû tíse-ʔsû áʔi=yi=ja máʔkaen fíʔthi-pa ávû kíniʔjin=ma
ANA-SML-DS =3 3SG-ATTR person=EXCL=CNTR somehow kill-SS fish tree=ACC
júkhaningae tsún-ña-ʔchu-ve-ja tsún-ʔfa
for later do-IRR-EN-ACC2-CNTR do-PLS
"Because of that, those very same ones had to kill, ensuring the survival of the fish tree." (Blaser and Chica Umenda, 2008, p. 25; `2024-06-07(1)_mll`)

d. ADDITIVE FOCUS *-ʔKHE* ADD + NEW TOPIC *-TA* NEW

ké=ʔkhe=ta =ti=ki tsa áʔi=ma shúndu-ʔsû-mbi?
2SG=ADD=NEW =YNQ=2 ANA person=ACC accompany-SN-NEG
"Art not thou also one of this man's disciples?"
(John 18:17; `2024-06-07(1)_mll`)

1ST ↓ / 2ND →	*-yi* EXCL	*-ʔkhe* ADD	*-ta* NEW	*-ja* CNTR
-yi- EXCL	—	*-yiʔkhe*	*-yita*	*-yija*
-ʔkhe- ADD	✗	—	*-ʔkheta*	✗
-ta- NEW	✗	✗	—	✗
-ja- CNTR	✗	✗	✗	—

Table 1: Co-occurrence among the A'ingae information structural markers

When it comes to word order, there is no particular position that the IS-marked constituents must occupy. For example, while new topics are often the first constituent in a sentence (9b, 9d), non-initial topics, including new topics (10a) and contrastive topics (10b), are also grammatical and found in natural corpora. Multiple IS-marked constituents (including constituents carrying the same marker) may appear per clause (9c, 10).

(10) Multiple topic per clause

a. MULTIPLE NEW TOPICS *-TA* NEW

ké?i=ta =ti=ki séje?pa=ve=ta á?mbian-?fa?
2PL=NEW =YNQ=2 poison=ACC2=NEW have-PLS

"Do you have poison?" (Quenamá, 1992, pp. 21, 57; `2024-06-10(1)_mll`)

b. MULTIPLE CONTRASTIVE TOPICS *-JA* CNTR

tayúpi=ja jungáesû=ma séma-?jen-?-jan tsé-?thi=nga =tsû má?kaen
long ago=CNTR something=ACC work-IPFV-T-CNTR STA-PLC=DAT =3 somehow
jungáesû=ma khá?na-sa?ne-jan íyû?û-pa kúndase-?fa
something=ACC steal-APPR-CNTR scold-SS tell-PLS

"In the old days, when (young people) were working on something (for someone else), (their fathers) would caution them not to steal anything from there." (Criollo and Blanco Pisabarro, 1992, pp. 43, 90; `2024-03-04(1)_sia`)

A'ingae topic markers interact with second-position clitics. A'ingae has five sentence-level clitics which encode the person feature of the matrix subject (first person *=ngi* 1, second person *=ki* 2, third person *=tsû* 3), evidentiality (reportative *=te* RPRT), and illocutionary force (polar interrogative *=ti* YNQ). The clitics are prototypically optional in matrix clauses, disallowed in subordinate clauses, and—when present—linearized after the first clausal constituent, i. e. in the second position. (These properties of the A'ingae second-position (P2) clitics can be derived by analyzing them as matrix-clausal C-heads. For more on the P2 clitics, see Dąbkowski (t.a.).)

However, the IS markers may cause deviations from those prototypical patterns. For example, although otherwise generally optional, a second-position clitic is preferred when a constituent marked with the new topic *-ta* NEW is present (11).

(11) P2 clitic preferred after new topic *-TA* NEW

a. NEW TOPIC WITHOUT PREGLOTTALIZATION *-TA*

áin=nda ?(*=tsû*) *tsái-ye atésû*
dog=NEW =3 bite-INF know

"The dog bites." (`2024-05-27(2)_sia`)

b. NEW TOPIC WITH PREGLOTTALIZATION *-?TA*

án-?-nda ?(*=ngi*) *khipué?sû-ya-mbi*
eat-T-NEW =1 hungry-IRR-NEG

"If I eat, I won't be hungry." (`2024-05-27(2)_sia`)

Moreover, the P2 clitics are ungrammatical immediately after constituents marked with the contrastive topic *-ja* CNTR (Fischer and Hengeveld, 2023) (12). These patterns hold regardless of whether (11b, 12b) or not (11a, 12a) the IS morpheme is preceded by the glottal stop *-?* T.

(12) P2 CLITIC DISALLOWED AFTER CONTRASTIVE TOPIC *-JA* CNTR

a. CONTRASTIVE TOPIC WITHOUT PREGLOTTALIZATION *-JA*

**áʔtse=ja*	(*=*tsû*)	*tsáʔu=nga*	(=*tsû*)	*káʔni*
hummingbird=CNTR	=3	house=DAT	=3	enter

"A hummingbird entered the house." (2024-04-03(1)_sia)

b. CONTRASTIVE TOPIC WITH PREGLOTTALIZATION *-ʔJA*

afa-khú-ʔ-ja	(*=*ngi*)	*sumbú-ya*	(=*ngi*)
speak-RCPR-T-CNTR	=1	leave-IRR	=1

"If I argue, I will leave." (2024-06-11(2)_mll)

4 A PRIMER ON PREDICATES

Most of the core data in Section 5 will pertain to the realization or non-realization of *-ʔ* T on predicates. As such, in this section, I present the basics of A'ingae predicate morphosyntax. Section 4.1 focuses on verbal predicates, including serial verb constructions. Section 4.2 briefly considers nominal predicates.

4.1 *Verbal predicates*

A'ingae predicates can vary greatly in morphological complexity. On one end, the head of a finite TP may consist of a bare root. On the other end, a plethora of grammatical categories can be expressed on a verb by means of suffixation (Dąbkowski, 2021b, 2024b). An example of a highly inflected verb is presented in (13).

(13) VERBAL PREDICATE STRUCTURE

[[[[[*kufi* *-án* $]_{\text{VceP}}$ *-ʔjen* *-ngi* $]_{\text{AspP}}$ *-ʔfa* *-mbi* $]_{\text{TP}}$ *-ʔni* *-nda* $]_{\text{CP}}$

play -CAUS -IPFV -VEN -PLS -NEG -IF.DS -NEW

"now$_{\text{NEW}}$, if$_{\text{IF}}$ (they$_{\text{PLS}}$) do not$_{\text{NEG}}$ come$_{\text{VEN}}$ to be$_{\text{IPFV}}$ making$_{\text{CAUS}}$ play, (someone else$_{\text{DS}}$) ..." (2024-06-18(1)_mll)

The morphological template of the A'ingae verb is given in Table 2.[9] The verbal root is at the bottom, and the subsequent morphosyntactic slots appear above the root, mimicking the orientation of a syntax tree. The A'ingae suffixes are grouped into four major functional projection domains. First, the voice domain (VceP) includes the causative *-ña*/*-an*/*-en* CAUS (i), reciprocal *-khu* RCPR (ii), and passive *-ye* PASS (iii).[10]

Second, the verbal inflectional domain (AspP) encompasses the aspectual suffixes (iv), including the imperfective *-ʔje* IPFV, the ingressive *-ji* INGR, the verbal diminutive *-kha* VDM,

9 Although I provide a template which illustrates the order of the morphemes in A'ingae verbal predicates, I do not view it as a theoretical primitive. Rather, I assume that suffix ordering is a consequence of selectional restrictions among functional heads, and I use the template as a convenient visual aid. A limitation of the templatic presentation will be apparent in Section 5.4, as non-final serialized verbs can host VceP, AspP,

CLAUSE-LEVEL SUFFIXES (CP)		
	(xv)	SUBJECT PERSON: *=ngi* 1, *=ki* 2, *=tsû* 3
	(xiv)	SENTENCE-LEVEL: *=te* RPRT, *=ti* YNQ
	(xiii)	TOPIC: *-ta* NEW, *-ja* CNTR
	(xii)	ADDITIVITY: *-ʔkhe* ADD
	(xi)	EXCLUSIVITY: *-yi* EXCL
	(x)	CLAUSE TYPE
		MATRIX: *-ja* IMP, *-kha* AIMP, *-ʔse* PERM, *-jama* PROH, *-ʔya* ASSR
		COSUBORDINATE: *-pa* SS, *-si* DS
		SUBORDINATE: *-saʔne* APPR, *-khen* TENT, (*-ʔa* IF.SS,) *-ʔni* IF.DS, *-ʔma* FRST
SITUATION-LEVEL SUFFIXES (TP)		
	(ix)	TENSE: (*-ʔ* T)
	(viii)	POLARITY: *-mbi* NEG, *-mbe* ADV.NEG
	(vii)	REALITY/FINITENESS: *-ya* IRR, *-ye* INF
	(vi)	SUBJECT NUMBER: *-ʔfa* PLS
VERBAL INFLECTIONAL SUFFIXES (AspP)		
	(v)	ASSOC MOTION: *-ʔngi* VEN, *-ʔnga* AND
	(iv)	ASPECT: *-ʔje* IPFV, *-ji* INGR, *-kha* VDM, *-ʔñakha* ITER
VOICE SUFFIXES (VceP)		
	(iii)	PASSIVE: *-ye* PASS
	(ii)	RECIPROCAL: *-khu* RCPR
	(i)	CAUSATIVE: *-ña*/*-an*/*-en* CAUS
VERBAL ROOT (√P)		
	(0)	VERBAL ROOT: √

Table 2: Morphological template of the A'ingae verb (building on Dąbkowski, 2024b, 2025c,d)

and the iterative *-ʔñakha* ITER, as well as the associated motion suffixes (v), including the venitive *-ʔngi* VEN and the andative *-ʔnga* AND.

and IS morphology, but not TP or clause-type morphology. For further discussion of templates as emergent, see e. g. Crippen (2019).

10 Some previous descriptions of the A'ingae verbal template (e. g. Dąbkowski, 2024b, in prep.) single out the causative morpheme *-ña*/*-an*/*-en* CAUS as the head of a separate verbal categorizing projection *v*P, and lump the other two voices (*-khu* RCPR, *-ye* PASS) with the aspectual suffixes under AspP. Dąbkowski's (2024b, in prep.) presentation has a phonological motivation—there is phonological evidence that *v*P is a phase, but there is no comparable evidence for VceP being a phase. Since the present paper is concerned only with morphosyntax, not phonology, I have opted for a more semantically transparent presentation of the verbal template.

Third, the situation-level domain (TP) is comprised of markers for (vi) subject number (plural *-ʔfa* PLS), (vii) reality (irrealis *-ya* IRR), finiteness (infinitive *-ye* INF), and (viii) polarity (negative *-mbi* NEG and negative adverbial *-mbe* ADV.NEG). A'ingae has no overt verbal morphology devoted to encoding tense. Nonetheless, following Shlonsky (1997, p. 3), I assume that every clause—by definition—contains the TP layer, so TP is universally present in all languages, including A'ingae. (For semantic evidence supporting the existence of TP in A'ingae, see Dąbkowski, in prep.) In Section 6, I will propose that the glottal stop under investigation *-ʔ* is in fact a spell-out of the T-head when linearly adjacent to an IS marker.

Fourth, the clause-level domain (CP) hosts suffixes that can be categorized as appearing on either matrix clauses (*-ja* IMP, *-kha* AIMP, *-ʔse* PERM, *-jama* PROH, *-ʔya* ASSR), cosubordinate clauses (*-pa* SS, *-si* DS), or subordinate clauses (*-saʔne* APPR, *-khen* TENT, *-ʔa* IF.SS, *-ʔni* IF.DS, *-ʔma* FRST).

A'ingae subordinate and cosubordinate clauses can be inflected for features such as subject plurality (*-ʔfa* PLS; 14a-14b), reality status (irrealis *-ya* IRR; 14a), polarity (negative *-mbi* NEG; 14a), and finiteness (infinitive *-ye* INF; 14b). Below, these morphemes are underlined. This shows that subordinate clauses consist of at least the TP layer (and also may have an overt or phonologically null CP layer).

(14) SUBORDINATE CLAUSE INFLECTION

a. FINITE SUBORDINATE CLAUSE

kúfe-ʔje-<u>ʔfa-ya-mbi</u>-ʔma =tsû avûjá-tshi-ʔfa-ya
play-IPFV-PLS-IRR-FRST =3 rejoice-ADJ-PLS-IRR
'Even if they are not playing, they will be happy." (2025-01-20(1)_mll)

b. INFINITIVE SUBORDINATE CLAUSE

mandá-ʔfa =tsû sumbú-<u>ʔfa-ye</u>
order-PLS =3 leave-PLS-INF
"They told them to leave." (2025-01-20(1)_mll)

Cosubordinate and subordinate clauses can then additionally be marked for discourse information with the new topic *-ta* NEW, contrastive topic *-ja* CNTR, exclusive focus *-yi* EXCL, and/or additive focus *-ʔkhe* ADD. I assume that the clause-type markers (x) are C-heads, and the discourse morphemes (xi-xiii) are adjuncts that can attach to different syntactic categories without changing the category label. Thus, the discourse markers attach to CPs, while also projecting a CP.

If appearing at the beginning of a sentence, most verbs can then be followed by second-position clitics (xiv-xv), including the reportative *=te* RPRT, the polar interrogative *=ti* YNQ, as well as the first *=ngi* 1, second *=ki* 2, or third *=tsû* 3 person subject clitic. (Since second-position clitics are phonologically integrated with the verb (Dąbkowski, 2024b), they are included here, despite technically not being a part of the verb's morphological template.)

In Table 2, the morphemes given in parentheses ( ) are overt feature-bundle realizations observed in specific syntactic contexts (to be discussed in Section 5). In other environments, those same feature bundles are phonologically null. Also note that Table 2 captures the relative order of verbal morphemes, but does not represent additional co-occurrence restrictions. For co-occurrence restrictions among the discourse markers, see Table 1. For co-occurrence restrictions among the TP morphemes, see Section 6.

4.1.1 *Verb serialization*

Serial verbs are monoclausal constructions that consist of two or more verbs without any overt subordinating, coordinating, or otherwise linking morphology (Aikhenvald, 2018; Lovestrand, 2021). Prototypically, all verbs within a serial verb construction (SVC) describe one event (Aikhenvald, 2018).

A'ingae SVCs can express a variety of complex events, as long as a discourse-cohesive interpretation can be supplied (Kalpande, 2024). Attested relations among individual verbs in a serial construction include *narration*, where each subsequent verb presents a temporally posterior event (15a), *result*, which combines a cause and an effect (15b), *continuation*, where the serialized verbs present different aspects of one event (15c), and *elaboration*, where the first verb introduces an event and the following verbs supply additional details (15d). For more on the A'ingae serial verb constructions and an analysis thereof couched within Discourse Coherence Theory (DCT) (e. g. Asher and Lascarides, 2003; Jasinskaja and Karagjosova, 2020; Kehler, 2001), see Kalpande (2024).

(15) DISCOURSE COHERENCE RELATIONS IN VERB SERIALIZATION

a. NARRATION RELATION
tíse =tsû khû́i jángi já
3SG =3 lie down get up go
"S/he lied down, got up, and left." (2025-08-28(1)_mll)

b. RESULT RELATION
tshéʔtshe fíʔthi-ʔya tise pû́she=ma
beat kill-ASSR 3SG wife=ACC
"He beat his wife to death." (Kalpande, 2024, p. 49; AnderBois and Silva, 2018, 20170807_flor_flanca_chiste_RJCL, 1:25; 2025-08-28(1)_mll)

c. CONTINUATION RELATION
tsún-mba jápa sétha jángi kû́ʔi-pa án
do-SS go-SS chant get up drink-SS eat
"Then they (held an ayahuasca ceremony, i. e. repeatedly) chanted, got up, and drank, and (then) ate." (Kalpande, 2024, p. 49; AnderBois and Silva, 2018, 20170731_yaje3_MM, 0:35; 2025-08-28(1)_mll)

d. ELABORATION RELATION
tíse =tsû tsû́i já
3SG =3 walk come
"S/he came walking." (2025-08-28(1)_mll)

Within a verb series, the non-terminal verbs can be inflected with voice suffixes, e. g. the reciprocal *-khu* RCPR (16a) and verbal inflectional suffixes, e. g. the imperfective *-ʔje* IPFV (16b-c).

(16) VOICE/VERBAL INFLECTION AND SVCS

a. RECIPROCAL NON-TERMINAL
tíseʔpa afá-khu jí-ʔfa
3SG speak-RCPR come-PLS
"They came arguing." (2025-09-05(1)_mll)

b. IMPERFECTIVE NON-TERMINAL
káni kúse =te=ngi ñá ána-ʔjen tsû́i
yesterday night =RPRT=1 1SG sleep-IPFV walk
"Last night I sleepwalked (I am told)." (2024-03-28(1)_sia)

c. IMPERFECTIVE NON-TERMINAL
tsún-si tsé-ne-jan fû́ndu-ʔje jí-ña
do-DS STA-ELAT-CNTR scream-IPFV come-PRSP
"Then he was coming from there screaming."
(Chica Umenda, 1980, p. 28; 2024-04-03(1)_sia)

Situation-level inflection, such as the negative *-mbi* NEG (17), and clause-type inflection, such as the prohibitive *-jama* PROH (18), are ungrammatical on non-terminal verbs in SVCs (17b, 18b), i. e. they may appear only on the last verb (17a, 18a). Situation-level and clause-type inflectional morphology always scopes over all the verbs within an SVC (17a, 18a).

(17) SITUATION-LEVEL INFLECTION AND SVCS

a. *-MBI* NEG SCOPES OVER ENTIRE SVC
tíse =tsû fû́ndu-ʔje jí-mbi
3SG =3 scream-IPFV come-NEG
"(S/he) didn't come screaming."
not: "*(S/he) was screaming (but) didn't come." (2025-08-28(1)_mll)

b. *-MBI* NEG UNGRAMMATICAL ON NON-TERMINAL VERB
**tíse =tsû fû́ndu-ʔje-mbi jí*
3SG =3 scream-IPFV-NEG come
intended: "(S/he) came without screaming." (2025-08-28(1)_mll)

(18) CLAUSE-TYPE INFLECTION AND SVCS

a. -JAMA PROH SCOPES OVER ENTIRE SVC

fû̂ndu-ʔje jí-jama

scream-IPFV come-PROH

"Do not come screaming."

not: "*(You)'re screaming (but) do not come." (2025-08-28(1)_mll)

b. -JAMA PROH UNGRAMMATICAL ON NON-TERMINAL VERB

**fû̂ndu-ʔje-jama jí*

scream-IPFV-PROH come

intended: "Come without screaming." (2025-08-28(1)_mll)

In short, non-final verbs can host voice (VceP) and verbal inflectional (AspP) morphology, but not situation-level (TP) or clause-type (CP) morphology. As such, we see that non-final verbs in a verb series are maximally AspPs.

4.2 *Non-verbal predicates*

Finally, I touch on the morphosyntax of non-verbal predicates. The first two projections (VceP and AspP) shown in Table 2 are exclusive to morphological verbs. This is to say, the voice, aspect, and associated motion suffixes can appear only on verbal heads.

The morphemes hosted by the latter two projections (TP and CP), on the other hand, can occur on both verbal and non-verbal predicates, including noun phrases and adjectives.[11] A'ingae lacks a copula verb; as such, the TP and CP morphemes attach directly to non-verbal predicates. An example of a noun phrase (with a bare possessor and the inflectional plural clitic *=ndekhû* PL.ANIM) functioning as a predicate is given in (19). IS morphemes behave in similar ways on verbal and non-verbal predicates. As such, in the rest of the paper, I group and discuss them together.

(19) NOMINAL PREDICATE STRUCTURE

tíseʔpa [[[*ña yayá =ndekhû* $]_{DP}$ *=ʔfa =mbi* $]_{TP}$ *=ʔni =nda* $]_{CP}$

3PL 1SG dad =PL.ANIM =PLS =NEG =IF.DS =NEW

"now$_{\text{NEW}}$, if$_{\text{IF}}$ they$_{\text{3PL}}$ are$_{\text{PLS}}$ not$_{\text{NEG}}$ my$_{\text{1SG}}$ parents$_{\text{PL.ANIM}}$, (someone else$_{\text{DS}}$) ... "

(2024-06-18(1)_mll)

5 THE DISTRIBUTION OF -ʔ T

The glottal stop under investigation *-ʔ* T appears only before discourse markers and only in certain morphosyntactic contexts. As such, in this section, I present and discuss the discourse markers as attached to constituents of various categories, and document in which configurations the glottal stop *-ʔ* T precedes them. First, I briefly go over non-predicates (§5.1). Then, I look at subordinate predicates of various types (§5.2–6).

5.1 *Non-predicates*

The A'ingae IS morphemes are most often observed on arguments and adjuncts of various kinds, including e. g. bare nouns (20a, 20h), pronouns (20b), question words (20c), stand-alone adjectives functioning as headless noun phrases (20d), nouns with inflectional clitics (20e), case-marked phrases (20f), and adverbs (20g).

(20) PLAIN REALIZATION ON NON-PREDICATES

a. BARE NOUN + NEW TOPIC -*TA* NEW

ánde=ta =tsû aʔí=ndekhû kánse-ʔchu túyaʔkaen kukuyá=ndekhû kánse-ʔchu
land=NEW =3 person=PL.ANIM live-EN and demon=PL.ANIM live-EN
"The earth is where human beings live, but also where the demons live."
(Blaser and Chica Umenda, 2008, p. 28; `2024-06-07(2)_mll`)

b. PRONOUN + CONTRASTIVE TOPIC -*JA* CNTR

tíse=ja ñuáʔme ke pûshe =tsû
3SG=CNTR truly 2SG wife =3
"Behold, of a surety she is thy wife." (Genesis 26:9; `2024-06-10(1)_mll`)

c. WH-PHRASE + EXCLUSIVE FOCUS -*YI* EXCL

junguésû=yi =tsû náʔen=ni kánse?
what=EXCL =3 river=LOC live
"What is it that lives in the river?" (`2024-06-10(1)_mll`)

d. ADJECTIVE + NEW TOPIC -*TA* NEW

áʔta-tshi-a=ta =ti fáeʔngae sínthia=iʔkhû kánʔjen-ña?
day-ADJ-ADN=NEW =YNQ together darkness=INS be.ANIM-IRR
"And what communion hath light with darkness?"
(2 Corinthians 6:14; `2024-06-10(1)_mll`)

e. INFLECTED NOUN + CONTRASTIVE TOPIC -*JA* CNTR

májan =tsû cofán=ndekhû=ja?
who =3 Cofán=PL.ANIM=CNTR
"Who are the Cofán?"
(Blaser and Chica Umenda, 2008, p. 10; `2024-06-10(1)_mll`)

f. CASE-MARKED PHRASE + EXCLUSIVE FOCUS -*YI* EXCL

áʔi=nga enthínge=ve=yi áfe
person=DAT half=ACC2=EXCL give
"(He) gave the (other) half to the man."
(Borman and Criollo, 1990, p. 87; `2024-06-10(1)_mll`)

11 A'ingae word order is largely free in matrix clauses, but predominantly verb-final in subordinate clauses. Consequently, I talk about the IS morphemes variably as attaching to clauses, predicates, or verbs/noun phrases/adjectives without any intended difference in meaning.

g. ADVERB + NEW TOPIC -*TA* NEW
tayúpi-ta =tsû Erisión tsáiʔmbi-ʔtshi teteté=ndekhû=ve fíthi~ʔthi
long ago-NEW =3 Erisión many-ADJ Waorani=PL.ANIM=ACC2 kill~PLA
"Once upon a time, Erisión killed many Waoranis."
(Blaser and Chica Umenda, 2008, p. 152; `2024-06-07(1)_mll`)

h. BARE NOUN + ADDITIVE FOCUS -*ʔKHE* ADD
máki máma=ʔkhe yáya=iʔkhû nasípa=ni já
some day mom=ADD dad=INS field=LOC go
"Some days, mom also goes with dad into the field."
(Borman and Chica Umenda, 1982b, p. 17; `2024-06-10(1)_mll`)

For all the different base of attachment categories seen above, and regardless of any potential functional morphology preceding the discourse morpheme, the new topic *-ta* NEW (20a, 20d, 20g), contrastive topic *-ja* CNTR (20b, 20e), and exclusive focus *-yi* EXCL (20c, 20f) are realized as plain (i. e. not preglottalized). The additive focus is *-ʔkhe* ADD is always preglottalized (20h).

In the rest of this section, I present the patterns of -CV/-ʔCV alternation when the IS markers attach to various verbs/predicates, including infinitive clauses (§5.2), finite clauses (with verbal and non-verbal predicates) (§5.3), serialized verbs (§5.4), nominalized clauses (§5.5), and adverbialized clauses (§5.6). These constitute the key data that will be accounted for in Section 6.

5.2 *Infinitive predicates*

First, I consider various uses of infinitive predicates. Infinitive clauses may, for example, function as subjects of stative predicates (21a), arguments selected by verbs such as the habitual auxiliary *atesû* 'know' (21b) or *seʔpi* 'forbid' (21g), same-subject rationale clauses (21c, 21d, 21h), different-subject rationale clauses (optionally introduced with the different-subject purpose subordinator *kûintsû/kaentsû* PURP.DS) (21e), or complements of the hortative operator *jinge(sû)* HORT (21f).

(21) PREGLOTTALIZED REALIZATION ON INFINITIVES WITH -*YE* INF

a. *YE*-SUBJECT + -ʔ T + NEW TOPIC -*TA* NEW
án-ñe-ʔ-nda =tsû injénge-ʔchu
eat-INF-T-NEW =3 needed-EN
"Eating is important." (`2024-05-27(2)_sia`)

b. *YE*-ARGUMENT + -ʔ T + CONTRASTIVE TOPIC -*JA* CNTR
atésû =ngi guáʔthi-an-ñe-ʔ-jan
know =1 boil-CAUS-INF-T-CNTR
"I (habitually) boil." (`2024-06-11(2)_mll`)

c. SAME-SUBJECT *YE*-RATIONALE + -ʔ T + EXCLUSIVE FOCUS *-YI* EXCL + NEW TOPIC *-TA* NEW
júsû afa-khú-ye-ʔ-yi-ta =ngi áʔingae=ma atésû-ʔje
only speak-RCPR-INF-T-EXCL-NEW =1 A'ingae=ACC learn-IPFV
"I am studying A'ingae only so that I can argue (with people)."
(2025-01-20(1)_mll)

d. SAME-SUBJECT *YE*-RATIONALE + -ʔ T + *-YI* EXCL + *-JA* CNTR
jí =ngi afa-khú-ye-ʔ-yi-ja
come =1 speak-RCPR-INF-T-EXCL-CNTR
"I only came (here) to argue." (2025-01-20(1)_mll)

e. DIFFERENT-SUBJECT *YE*-RATIONALE + -ʔ T + CONTRASTIVE TOPIC *-JA* CNTR
tíse =tsû avû́ja-en (kû́intsû) kúfe-ʔje-ye-ʔ-ja
3SG =3 rejoice-CAUS PURP.DS play-IPFV-INF-T-CNTR
"S/he encouraged them to be playing." (2024-06-10(1)_mll)

f. *YE*-HORTATIVE + -ʔ T + EXCLUSIVE FOCUS *-YI* EXCL
jingésû kíʔan-ʔfa-ye-ʔ-yi
HORT strong-PLS-INF-T-EXCL
"Let's just be strong!" (2024-06-25(1)_mll)

g. *YE*-ARGUMENT + ADDITIVE FOCUS *-ʔKHE* ADD
án-ñe-ʔkhe séʔpi =ngi
eat-INF-ADD forbid =1
"I also forbid eating." (2024-06-11(2)_mll)

h. SAME-SUBJECT *YE*-RATIONALE + ADDITIVE FOCUS *-ʔKHE* ADD+ NEW TOPIC *-TA* NEW
afa-khú-ye-ʔkhe-ta =ngi áʔingae=ma atésû-ʔje
speak-RCPR-INF-ADD-NEW =1 A'ingae=ACC learn-IPFV
"I am studying A'ingae also so that I can argue (with people)."
(2025-01-20(1)_mll)

All of these uses are compatible with IS markers. When appearing on an infinitive verb, regardless of its function,[12] the new topic *-ta* NEW (21a), contrastive topic *-ja* CNTR (21b, 21e), and exclusive focus *-yi* EXCL (21c, 21d, 21f) are always preceded by the glottal stop *-ʔ*. The additive focus *-ʔkhe* ADD (21g, 21h) is always realized as preglottalized.

The presence of earlier inflectional morphology appearing to the left of *-ye* INF, including VceP (21b, 21c, 21d, 21h), AspP (21e), and TP (21f) suffixes, does not affect the the presence of the glottal stop *-ʔ*. If multiple IS markers appear on a predicate (such as an infinitive), a glottal stop is observed only before the first one (21c, 21d, 21h). (This fact strengthens the argument against preglottalization being analyzed as part some kind of allomorph of each of the IS suffixes.)

12 Following standard assumptions (e. g. Adger, 2003), some of the uses of infinitive clauses may be analyzed as TPs, while others as CPs. Since their specific syntactic category will turn out immaterial in Section 6, I abstract away from this distinction here.

5.3 *Finite predicates*

Now, I will look at finite predicates with IS morphemes. A'ingae finite clauses may be (co)subordinated by one of the following morphemes: the same-subject marker *-pa* SS (§5.3.1), different-subject marker *-si* DS (§5.3.2), apprehensional *-saʔne* APPR (§5.3.3), tentative *-khen* TENT (§5.3.4), same-subject conditional antecedent marker realized as *-ʔa* IF.SS before *-ʔkhe* ADD and as phonologically null otherwise (§5.3.5), different-subject conditional antecedent marker *-ʔni* IF.DS (§5.3.6), and frustrative *-ʔma* FRST. The first six of these subordinating morphemes may be followed by discourse markers.

5.3.1 *Same-subject*

A'ingae has two switch-reference morphemes that encode whether or not the subjects of two connected clauses are the same (same-subject *-pa* SS and different-subject *-si* DS), and appear in three constructions: clause chaining, adverbial clauses, and "bridging" clause linkage (AnderBois, Altshuler, and Silva, 2023). The adverbial clauses, often (though not always) translated with "because," are compatible with IS markers.

The same-subject *-pa* SS is used if the subject of the *pa*-marked clause and the subject of the following clause are the same (22). When appearing on a same-subject clause, the new topic *-ta* NEW (22b), contrastive topic *-ja* CNTR (22c), and exclusive focus *-yi* EXCL (22d) are not preceded by the glottal stop. The additive focus *-ʔkhe* ADD (22e) is always preglottalized.

(22) PLAIN REALIZATION ON SAME-SUBJECT CLAUSES WITH *-PA* SS

a. *PA*-PREDICATE

ápi=ma píʔpi-pa ísû
clay=ACC mold-SS take

"(S/he) made a clay pot and picked it up."
(Borman and Chica Umenda, 1977, p. 13; `2024-02-20(1)_sia`)

b. *PA*-PREDICATE + NEW TOPIC *-TA* NEW

amphí-pa-ta =ti=ki tsífu=ja báthi-ʔchu-mbi?
fall-SS-NEW =YNQ=2 neck=CNTR dislocate-EN-NEG

"Didn't you injure your neck by falling?"
(Borman, Cooper, and Criollo, 1991, p. 60; `2024-06-07(1)_mll`; `2025-07-17(1)_mll`)

c. *PA*-PREDICATE + CONTRASTIVE TOPIC *-JA* CNTR

afé-ya =ti fae regálo=ve ké=nga ke yáya-pa-ja?
give-IRR =YNQ one gift=ACC2 2SG=DAT 2SG dad-SS-CNTR

"Will he give you a gife because he's your dad?" (`2024-06-10(1)_mll`)

d. *PA*-PREDICATE + EXCLUSIVE FOCUS *-YI* EXCL
túya án-ʔjen-mba-yi =ki jí-mbi?
still eat-IPFV-SS-EXCL =2 come-NEG
"You didn't come just because you were still eating?" (2024-06-10(1)_mll)

e. *PA*-PREDICATE + ADDITIVE FOCUS *-ʔKHE* ADD
amphí-pa-ʔkhe =ti=ki tsífu=ja báthi-ʔchu-mbi?
fall-SS-ADD =YNQ=2 neck=CNTR dislocate-EN-NEG
"Did you injure your neck also by falling?" (2024-06-06(1)_mll)

5.3.2 *Different-subject*

The different-subject *-si* DS is used if the subject of the *si*-marked clause and the subject of the following clause are different (23). When appearing on a different-subject clause, the new topic *-ta* NEW (23b), contrastive topic *-ja* CNTR (23c), and exclusive focus *-yi* EXCL (23d) are not preglottalized.

(23) PLAIN REALIZATION ON DIFFERENT-SUBJECT CLAUSES WITH *-SI* DS

a. *SI*-PREDICATE
arápa pá-si =tsû ísû
chicken die-DS =3 take
"The chicken died and s/he picked it up."
(Borman and Chica Umenda, 1977, p. 13; 2024-02-20(1)_sia)

b. *SI*-PREDICATE + NEW TOPIC *-TA* NEW
tíse ke yáya-si-ta =ti=ki isû-ya fae regálo=ma?
3SG 2SG dad-DS-NEW =YNQ=2 get-IRR one gift=ACC
"Will you get a gift from him because he's your dad?" (2024-06-10(1)_mll)

c. *SI*-PREDICATE + CONTRASTIVE TOPIC *-JA* CNTR
kéʔi atesû́-ʔfa-mbi-si-ja kéʔi=nga teváen-mbi =ngi
2PL know-PLS-NEG-DS-CNTR 2PL=DAT write-NEG =1
"I do not write to you because you do not know (the truth)."
(1 John 2:21; 2024-06-10(1)_mll)

d. *SI*-PREDICATE + EXCLUSIVE FOCUS *-YI* EXCL
tsá=ma áthe-pa kúse índi-ye rúnʔda-ʔma áthe-mbi tíse ána-si-yi
ANA=ACC see-SS night catch-INF wait-FRST see-NEG 3SG sleep-DS-EXCL
jí-pa já-ʔje-si
come-SS go-IPFV-DS
"Having seen her, (he) waited to catch (her) at night, but (he) didn't see (her), because (she) would only come and go when he was asleep."
(Blaser and Chica Umenda, 2008, p. 156; 2024-06-18(1)_mll; 2025-01-20(1)_mll)

e. SI-PREDICATE + ADDITIVE FOCUS -ʔKHE ADD
píyi kán-si-ʔkhe =ti íñe?
turn look-DS-ADD =YNQ hurt
"Does it hurt also when you turn (your head)?" (2024-06-06(1)_mll)

5.3.3 *Apprehensional*

The apprehensional *-saʔne* APPR marks undesirable potential future situations (24). Three main verbal functions of the morpheme may be distinguished: avertive precautioning (24a, 24c, 24e), in-case precautioning (24b), and *fear*-complementizer (24d) (AnderBois and Dąbkowski, 2020, 2025; Dąbkowski and AnderBois, 2024, t.a.). (These three functions of *-saʔne* APPR are similar to the range of uses of the somewhat archaic English *lest*; Dąbkowski and AnderBois, 2024.) When appearing on an apprehensional-marked clause, the new topic *-ta* NEW (24b), contrastive topic *-ja* CNTR (24c), and exclusive focus *-yi* EXCL (24d) surface without a preceding glottal stop.

(24) PLAIN REALIZATION ON APPREHENSIONAL CLAUSES WITH -SAʔNE APPR

a. SAʔNE-AVERTIVE PRECAUTIONING
sinthí-mbe-ʔ-yi tsû kán-ña-ʔchu kháse anjámpa tshán-saʔne
blow-ADV.NEG-T-EXCL =3 AUX-IRR-EN again blood bleed-APPR
"You should not blow your nose, so that you don't make it bleed again."
(Pederson and Cooper, 1982, pp. 68–69; 2024-02-20(1)_sia)

b. SAʔNE-IN-CASE PRECAUTIONING + NEW TOPIC -TA NEW
û̂njin tû̂i-saʔne-nda =ngi bûthú-ya
rain rain-APPR-NEW =1 run-IRR
"I will run in case it rains." (2024-06-18(1)_mll)

c. SAʔNE-AVERTIVE PRECAUTIONING + CONTRASTIVE TOPIC -JA CNTR
tayúpi=ja jungáesû=ma séma-ʔjen-ʔ-jan tsé-ʔthi=nga =tsû máʔkaen
long ago=CNTR something=ACC work-IPFV-T-CNTR STA-PLC=DAT =3 somehow
jungáesû=ma kháʔna-saʔne-jan íyûʔû-pa kúndase-ʔfa
something=ACC steal-APPR-CNTR scold-SS tell-PLS
"In the old days, when (young people) were working on something (for someone else), (their fathers) would caution them not to steal anything from there."
(Criollo and Blanco Pisabarro, 1992, pp. 43, 90; 2024-03-04(1)_sia)

d. SAʔNE-COMPLEMENT OF *FEAR* + EXCLUSIVE FOCUS -YI EXCL
tíseʔpa ña yayá-ndekhû-ʔfa-saʔne-ñi =ngi dyúju
3PL 1SG dad-PL.ANIM-PLS-APPR-EXCL =1 be afraid
"I fear only that they are my parents." (2024-06-11(1)_mll)

e. SAʔNE-AVERTIVE PRECAUTIONING + ADDITIVE FOCUS -ʔKHE ADD
índi-saʔne-ʔkhe =ngi bûthú-ya
catch-APPR-ADD =1 run-IRR
"I will run also so that (s/he) doesn't catch (me)." (2024-06-18(1)_mll)

5.3.4 *Tentative*

The tentative *-khen* TENT introduces subordinate verbs in auxiliary constructions that express an attempt or effort on the part of the subject to accomplish something (25). A *khen*-marked clause may be introduced as a complement to verbs such as *tsun* 'do' (25b, 25e), *iyikhu* 'fight' (25c), and *vana* 'suffer' (25d). They can often be translated with "try to (do sth)" or "struggle to (do sth)." When appearing on a tentative clause, the new topic *-ta* NEW (25b), contrastive topic *-ja* CNTR (25c), and exclusive focus *-yi* EXCL (25d) surface without preglottalization.

(25) PLAIN REALIZATION ON TENTATIVE CLAUSES WITH -*KHEN* TENT

a. *KHEN*-PREDICATE
seʔjé-khen =ngi tsún-ʔjen
cure-TENT =1 do-IPFV
"I am trying to cure." (2022-07-01(1)_jxm)

b. *KHEN*-PREDICATE + NEW TOPIC -*TA* NEW
athe-yé-khen-nda =ngi tsún-ʔjen
see-PASS-TENT-NEW =1 do-IPFV
"I am trying to be seen." (2024-06-11(2)_mll)

c. *KHEN*-PREDICATE + CONTRASTIVE TOPIC -*JA* CNTR
inʔján-khen-jan iyíkhu-ʔje =ngi
believe-TENT-CNTR fight-IPFV =1
"I am wrestling to believe." (2024-06-18(1)_mll)

d. *KHEN*-PREDICATE + EXCLUSIVE FOCUS -*YI* EXCL
iʔná-khen-ñi =ngi vána-ʔjen
cry-TENT-EXCL =1 suffer-IPFV
"I am struggling only to cry." (2025-07-17(1)_mll)

e. *KHEN*-PREDICATE + ADDITIVE FOCUS -*ʔKHE* ADD
athé-khen-ʔkhe =ngi tsún-ʔjen
see-TENT-ADD =1 do-IPFV
"I am also trying to see." (2024-06-18(1)_mll)

In an interim summary, so far we have seen that when an IS morpheme appears on a clause subordinated with the same-subject *-pa* SS (§5.3.1), different-subject *-si* DS (§5.3.2), apprehensional *-saʔne* APPR (§5.3.3), or tentative *-khen* TENT (§5.3.4), the IS morpheme is not preceded by a glottal stop *-ʔ*.

5.3.5 *Same-subject conditional*

Now, I move on to the A'ingae conditional constructions, where a different pattern is seen. The A'ingae conditionals may have afactual and factual interpretations. Afactual conditionals do not presuppose that the antecedent will obtain and can often be translated with "if."

Factual conditionals presuppose that the antecedent will obtain and can be translated with "when." There is no morphological distinction between afactual and factual conditionals.

There is, however, a morphological distinction that tracks whether the subject of the antecedent is the same or different from the subject of the consequent. As such, A'ingae conditionals participate in the same switch-reference system as that encoded by *-pa* SS and *-si* DS in non-conditional constructions.

Same-subject conditional antecedents do not, in most circumstances, receive any overt dedicated marking. Rather, they are distinguished by carrying only an IS marker. When introducing a same-subject (SS) conditional antecedent, the new topic *-ta* NEW (26a), contrastive topic *-ja* CNTR (26b), or exclusive focus *-yi* EXCL (26c), always appear with preceding preglottalization. If more than one discourse marker appears on an SS clause, preglottalization appears only before the first one (26d-e) (except for the additive focus *-ʔkhe* ADD (26f), which is always preglottalized). In most other constructions, the IS morphemes are optional (22a, 23a, 24a, 25a, 32a). However, an SS antecedent is always introduced by at least one IS morpheme (26).

(26) PREGLOTTALIZED REALIZATION ON SAME-SUBJECT CONDITIONALS

a. UNMARKED ANTECEDENT + -ʔ T + NEW TOPIC -TA NEW

ña	*yáya=ʔ=ta*	*=tsû*	*afé-ya*	*fae*	*regálo=ve*	*ña=nga*
1SG	dad=T=NEW	=3	give-IRR	one	gift=ACC2	1SG=DAT

"If (he)'s my dad, he'll give me a gift." (2024-06-10(1)_mll)

b. UNMARKED ANTECEDENT + -ʔ T + CONTRASTIVE TOPIC -JA CNTR

afa-khú-ʔ-ja	*sumbú-ya*	*=ngi*
speak-RCPR-T-CNTR	leave-IRR	=1

"If I argue, I will leave." (2024-06-11(2)_mll)

c. UNMARKED ANTECEDENT + -ʔ T + EXCLUSIVE FOCUS -YI EXCL

thési=ma	*afáse-ʔje-ʔ-yi*	*=ngi*	*bûthú-ya*
jaguar=ACC	criticize-IPFV-T-EXCL	=1	run-IRR

"I will run criticizing only the jaguar." (2024-06-06(2)_mll)

d. UNMARKED ANTECEDENT + -ʔ T + EXCLUSIVE FOCUS -YI EXCL + NEW TOPIC -TA NEW

tsá-ʔka-en	*kánse-ʔfa-ʔ-yi-ta*	*=ki*	*ñuáʔme*	*shaká-ʔfa-ya-mbi*
ANA-SML-ADV	live-PLS-T-EXCL-NEW	=2	truly	lack-PLS-IRR-NEG

"Only if you live like this, you will truly not lack (anything)."
(Thessalonians 4:12; 2024-06-07(1)_mll)

e. ANTECEDENT + -ʔ T + EXCLUSIVE FOCUS -YI EXCL + CONTRASTIVE TOPIC -JA CNTR

thési=ma	*áfase-ʔ-yi-ja*	*bûthú-ya*	*=ngi*
jaguar=ACC	criticize-T-EXCL-CNTR	run-IRR	=1

"Only if I criticize a jaguar, I will run." (2024-06-06(2)_mll)

f. UNMARKED ANTECEDENT + -ʔ T + EXCLUSIVE FOCUS -*YI* EXCL + -*ʔKHE* ADD
vá ni fae ávû=mbi=ʔ=yi=ʔkhe =tsû ami-án-ña-mbi
DEM neither one fish=NEG=T=EXCL=ADD =3 fill up-CAUS-IRR-NEG
"If this is not even a single fish, it won't fill me up." (2024-06-11(2)_mll)

(In natural corpora, same-subject antecedents are most frequently observed as marked with the preglottalized -*ʔta* (26a) or -*ʔja* (26b). As such, one may mistake -*ʔta* and -*ʔja* for morphemes directly expressing the same-subject conditional antecedent meaning. A fuller consideration of the distributional facts, however, reveals that both the glottal stop -*ʔ* and -*ta*/-*ja* appear independently in other contexts.)

A different structure is seen when the the additive focus marker -*ʔkhe* ADD appears on a same-subject conditional antecedent. In those cases, an additional morpheme -*ʔa* IF.SS is obligatorily present between the subordinated verb and -*ʔkhe* ADD (27).

(27) SAME-SUBJECT CONDITIONAL MARKED WITH -*ʔA* IF.SS WHEN FOLLOWED BY -*ʔKHE* ADD

a. *ʔA*-ANTECEDENT + ADDITIVE FOCUS -*ʔKHE* ADD
khúpa-ʔnga-pa jí-ʔa-ʔkhe =tsû utishí-ya-ʔchu
defecate-AND-SS come-IF.SS-ADD =3 wash hands-IRR-EN
"After using the restroom, one must also wash hands."
(Borman and Chica Umenda, 1982b, p. 27; 2024-06-07(1)_mll)

b. *ʔA*-ANTECEDENT + ADDITIVE FOCUS -*ʔKHE* ADD
kuénza áʔi=ve dá-ʔa-ʔkhe ñú-ʔa áʔi=ya =tsû
old person=ACC2 become-IF.SS-ADD good-ADN person=IRR =3
"Also when s/he grows old, s/he will be a good person."
(Borman, 1981, pp. 16–17; 2025-01-20(1)_mll)

The morpheme -*ʔa* does not (often) appear in circumstances other than before -*ʔkhe* ADD in same-subject conditional clauses. In addition, it is disallowed before the other IS markers (28). (Its use with the other IS markers is ungrammatical regardless of whether or not they are preceded by -*ʔ* T.) As such, in Section 6, I will analyze it as an allomorph of the (otherwise phonologically null) feature bundle [IF, SS] selected specifically when adjacent to -*ʔkhe* ADD.

(28) OVERT -*ʔA* IF.SS ALLOWED AND REQUIRED ONLY IMMEDIATELY BEFORE -*ʔKHE* ADD

a. **jí-ʔa(-ʔ)-ta*	b. **jí-ʔa(-ʔ)-ja*	c. **jí-ʔa(-ʔ)-yi*	d. **jí-ʔkhe*
come-IF.SS-T-NEW	come-IF.SS-T-CNTR	come-IF.SS-T-EXCL	come-ADD
intended: "if (sb) comes"	intended: "if (sb) comes"	intended: "only if (sb) comes"	intended: "also if (sb) comes"

(2025-01-20(1)_mll)

Recall that the IS morphemes may also appear on non-predicates, in which case they are not preceded by a glottal stop. This gives rise to minimal pairs, where non-verbal predicates

(29b, 30b, 31b) may be distinguished from arguments (29a, 30a, 31a) solely by the presence of glottalization before the IS marker.

(29) MINIMAL (ʔ)TA-PAIR ON A NOUN

a. ARGUMENT + NEW TOPIC -TA NEW
tíse chán=da =tsû jí-ya-mbi
3SG mother=NEW =3 come-IRR-NEG
"His/her mother will not come." (2025-01-20(1)_mll)

b. PREDICATE + -ʔ T + NEW TOPIC -TA NEW
tíse chán=ʔ=da =tsû jí-ya-mbi
3SG mother=T=NEW =3 come-IRR-NEG
"If she is a mother, she won't come." (2025-01-20(1)_mll)

(30) MINIMAL (ʔ)TA-PAIR ON AN ADJECTIVE

a. ARGUMENT + NEW TOPIC -TA NEW
ránde=ta =tsû jí-ya-mbi
large=NEW =3 come-IRR-NEG
"(The) large (one) will not come." (2025-01-20(1)_mll)

b. PREDICATE + -ʔ T + NEW TOPIC -TA NEW
ránde=ʔ=ta =tsû jí-ya-mbi
large=T=NEW =3 come-IRR-NEG
"If (s/he) is large, (s/he) will not come." (2024-06-20(2)_mll)

(31) MINIMAL (ʔ)JA-PAIR ON A NOUN

a. ARGUMENT + CONTRASTIVE TOPIC -JA CNTR
yáya=ja ñá=ma íyûʔû-ya =tsû
dad=CNTR 1SG=ACC scold-IRR =3
"(My) dad will advise me." (2024-06-20(2)_mll)

b. PREDICATE + -ʔ T + CONTRASTIVE TOPIC -JA CNTR
yáya=ʔ=ja ñá=ma íyûʔû-ya =tsû
dad=T=CNTR 1SG=ACC scold-IRR =3
"If (he)'s a dad, he'll advise me." (2024-06-20(2)_mll)

5.3.6 *Different-subject conditional*

Finally, different-subject conditional antecedents are introduced by the always overt *-ʔni* IF.DS (32).[13] Different-subject antecedents are frequently accompanied by an IS marker (32b-e), but that is not obligatory (32a). When appearing after *-ʔni* IF.DS, the new topic *-ta* NEW

13 In some previous descriptions and analyses, including AnderBois, Altshuler, and Silva's (2023), Fischer and Hengeveld's (2023), and Hengeveld and Fischer's (2018), the different-subject conditional antecedent *-ʔni* IF.DS was conflated with the locative case marker *=ni* LOC and glossed as such. In this paper, *-ʔni* IF.DS and *=ni* LOC are analyzed as two different morphemes. For an argument for supporting this distinction, see Dąbkowski (2025c, p. 120).

(32b), contrastive topic *-ja* CNTR (32c), and exclusive focus *-yi* EXCL (32d) surface as not preglottalized.

(32) PLAIN REALIZATION ON DIFFERENT-SUBJECT CONDITIONALS WITH *-ʔNI* IF.DS

a. *ʔNI*-ANTECEDENT

khén íngi sû̂-ʔje-ʔni fáeʔkhu-e-ta púʔtaen-ʔfa-ʔya
THUS 1PL say-IPFV-IF.DS one-ADV-NEW shoot-PLS-ASSR
"While we were saying this, there was one gunshot."
(Quenamá, 1992, pp. 25, 42; `2025-01-20(1)_mll`)

b. *ʔNI*-ANTECEDENT + NEW TOPIC *-TA* NEW

ñá náʔen=nga kháya-ʔje-ʔni-nda =tsû tíseʔpa séma-ʔjen-ʔfa
1SG river=DAT swim-IPFV-IF.DS-NEW =3 3PL work-IPFV-PLS
"When I am swimming in the river, they are working." (`2022-07-01(2)_sia`)

c. *ʔNI*-ANTECEDENT + CONTRASTIVE TOPIC *-JA* CNTR

ránde=ʔni=jan chavá-ya =ngi
large=IF.DS=CNTR buy-IRR =1
"If it's large, I'll buy it." (`2024-06-10(1)_mll`)

d. *ʔNI*-ANTECEDENT + EXCLUSIVE FOCUS *-YI* EXCL

thési=ma áthe-ʔfa-ʔni-ñi =tsû bû̂thu
jaguar=ACC see-PLS-IF.DS-EXCL =3 run
"As soon as they saw a jaguar, (another person) ran." (`2024-06-18(1)_mll`)

e. *ʔNI*-ANTECEDENT + ADDITIVE FOCUS *-ʔKHE* ADD

kúndase khû̂i, tíse dyú-ʔni-ʔkhe
tell lie down 3SG fear-IF.DS-ADD
"While lying (on the ground, the severed head) talked (to him/her), even though s/he was scared." (Chica Umenda, 1980, p. 26; `2024-03-04(1)_sia`)

In an interim summary, we have seen that a glottal stop -ʔ T appears on infinitival *ye*-marked clauses and morphologically unmarked same-subject antecedents when followed by a discourse marker. In other cases, no glottal stop is observed before the new topic *-ta* NEW, contrastive topic *-ja* CNTR, or exclusive focus *-yi* EXCL.

In the rest of this section, I consider three additional cases, where information structural markers attach to non-final verbs in SVCs (§5.4), nominalized clauses (§5.5), and adverbialized clauses (§5.6).

5.4 *Serialized verbs*

Serial verb constructions are predicates which comprise multiple verbs combined without any overt linking morphology (33). Non-final verbs in SVCs may be marked with information structural markers, including the new topic *-ta* NEW (33b, 33e), contrastive topic

-ja CNTR (33c), exclusive focus *-yi* EXCL (33d, 33e), and additive focus *-ʔkhe* ADD. When appearing on a non-final serialized verb, the IS markers surface without preglottalization.

(33) PLAIN REALIZATION ON NON-TERMINAL SERIALIZED VERBS

a. IMPERFECTIVE NON-TERMINAL VERB
tsún-si tsé-ne-jan fʉ̂ndu-ʔje jí-ña
do-DS STA-ELAT-CNTR scream-IPFV come-PRSP
"Then he was coming from there screaming."
(Chica Umenda, 1980, p. 28; `2024-04-03(1)_sia`)

b. NON-TERMINAL VERB + NEW TOPIC *-TA* NEW
fʉ̂ndu-ʔje-ta jí
sceam-IPFV-NEW come
"(S/he) came screaming." (`2025-08-28(1)_mll`)

c. NON-TERMINAL VERB + CONTRASTIVE TOPIC *-JA* CNTR
fʉ̂ndu-ʔje-ja jí
sceam-IPFV-CNTR come
"(S/he) came screaming." (`2025-08-28(1)_mll`)

d. NON-TERMINAL VERB + EXCLUSIVE FOCUS *-YI* EXCL
fʉ̂ndu-ʔje-yi jí
sceam-IPFV-EXCL come
"(S/he) came only screaming." (`2025-08-28(1)_mll`)

e. NON-TERMINAL VERB + EXCLUSIVE FOCUS *-YI* EXCL + NEW TOPIC *-TA* NEW
fʉ̂ndu-ʔje-yi-ta jí-ya-mbi
sceam-IPFV-EXCL-NEW come-IRR-NEW
"(S/he) will not come only screaming." (`2025-08-28(1)_mll`)

As we saw in Section 5.3.5, when attached to full predicates and preceded by -ʔ T, the information structural markers introduce same-subject conditionals. This gives rise to minimal pairs, where non-final terminal verbs in SVCs (34a) may be distinguished from SS conditional antecedents (34b) solely by the presence of glottalization before the IS marker.

(34) MINIMAL (ʔ)*YITA*-PAIR ON A VERB

a. NON-TERMINAL VERB + EXCLUSIVE FOCUS *-YI* EXCL + NEW TOPIC *-TA* NEW
fʉ̂ndu-ʔje-yi-ta jí-ya-mbi
sceam-IPFV-EXCL-NEW come-IRR-NEW
"(S/he) will not come only screaming." (`2025-08-28(1)_mll`)

b. SAME-SUBJECT ANTECEDENT + -ʔ T + EXCLUSIVE FOCUS *-YI* EXCL + NEW TOPIC *-TA* NEW
fʉ̂ndu-ʔje-ʔ-yi-ta jí-ya-mbi
sceam-IPFV-T-EXCL-NEW come-IRR-NEW
"(S/he) will not come only when (s/he)'s screaming." (`2025-08-28(1)_mll`)

5.5 *Nominalized predicates*

Next, I consider nominalized predicates. A'ingae has a large set of suffixes which can nominalize verbs and predicates of various structural height, giving rise to a variety of subordinate clauses (Dąbkowski, 2025b, in prep. Fischer and Hengeveld, 2023). This set of nominalizers includes, but is not limited to, the subject nominalizer *-ʔsû* SN (35a), negative individual nominalizer *-mbi* NIN (35b), lateral classifier *-ʔfa* LAT (35c), place classifier *-ʔthi* PLC (35d), bound space classifier *-khû* BND (35e), event nominalizer *-ʔchu* EN (35f, 35g), and others.

When the new topic *-ta* NEW (35a, 35e), contrastive topic *-ja* CNTR (35b, 35f), or exclusive focus *-yi* EXCL (35c, 35d) appear on a verb or clause subordinated with any of the nominalizers, they are not preceded by a glottal stop. The additive focus *-ʔkhe* ADD is always preglottalized (35g).

(35) PLAIN REALIZATION ON NOMINALIZATIONS

a. *ʔSÛ*-NOMINALIZATION + NEW TOPIC *-TA* NEW

ké=ma atesi-án-ʔsû-ta =ti Chíga éthi=ma ñuñá-ña-ʔchu impuésto=ma
2SG=ACC learn-CAUS-SN-NEW =YNQ God house=ACC make-IRR-EN tax=ACC
afepuén-ʔjen?
pay-IPFV
"'Does your teacher (not) pay the temple tax?"
(Matthew 17:24; `2025-07-17(1)_mll`)

b. *MBI*-NOMINALIZATION + CONTRASTIVE TOPIC *-JA* CNTR

tsún-si tsa panzá-mbi-ja tise ûfákhuʔkhu=ma isû́-pa já áʔchu=ma ûfá-ye
do-DS ANA hunt-NIN-CNTR 3SG blowgun=ACC take-SS go monkey=ACC blow-INF
"Then the bad hunter took his blowgun and went to kill the monkey."
(Blaser and Chica Umenda, 2008, p. 120; `2025-07-17(1)_mll`)

c. *ʔFA*-NOMINALIZATION + EXCLUSIVE FOCUS *-YI* EXCL

fae putáen-ʔfa-yi khuángiʔkhu-e fíʔthi thúfi=ma
one shoot-LAT-EXCL two-ADV kill parrot=ACC
"(I) killed two parrots (with) one shot."
(Borman and Chica Umenda, 1982b, p. 37; `2025-07-17(1)_mll`)

d. *ʔTHI*-NOMINALIZATION + EXCLUSIVE FOCUS *-YI* EXCL + ADDITIVE FOCUS *-ʔKHE* ADD

ni fae teváen-ʔthi-yi-ʔkhe pasá-ya-mbi
neither one write-PLC-EXCL-ADD pass-IRR-NEG
"One tittle (of the law) shall not fail." (Luke 16:17; `2025-07-17(1)_mll`)

e. *KHÛ*-NOMINALIZATION + NEW TOPIC *-TA* NEW

dû́ʔshû tise máma=me rúʔnda-ya-khû-ta =tsû ránde
child 3SG mom=ACC2 wait-IRR-BND-NEW =3 large
"The room where a child will wait for their mom is large." (`2025-07-17(1)_mll`)

f. ʔCHU-NOMINALIZATION + CONTRASTIVE TOPIC -JA CNTR
tsá=ʔma séfa=ni=ʔsû íngi túya áthe-mbi-ʔchu-a-ja tsángae =tsû jinchú-ya
ANA=FRST sky=LOC=ATTR 1PL still see-NEG-EN-ADN[14]-CNTR forever =3 remain-IRR
"But what is unseen in the heavens is eternal."
(2 Corinthians 4:18; `2025-07-17(1)_mll`)

g. ʔCHU-NOMINALIZATION + ADDITIVE FOCUS -ʔKHE ADD
tíse =tsû kínse-tshi túyaʔkaen tsúve ínjan-ʔjen-ʔchu-ʔkhe ñú-tshi
3SG =3 healthy-ADJ and head like-IPFV-EN-ADD good-ADJ
"S/he is healthy and mentally well." (`2025-07-22(1)_mll`)

When a nominalized clause functions as a predicate in a same-subject conditional antecedent, however, the discourse marker is preceded by a glottal stop *-ʔ* (36b). Such constructions may, again, contrast minimally with similar sentences where the nominalized clause is an argument and the glottal stop is not observed (36a).

(36) MINIMAL ʔCHU(ʔ)TA-PAIR

a. NOMINALIZED ʔCHU-ARGUMENT + NEW TOPIC -TA NEW
pá-ʔchu-ta =tsû jí-ya-mbi
die-EN-NEW =3 come-IRR-NEG
"(The) dead (one) will not come." (`2024-06-20(2)_mll`)

b. NOMINALIZED ʔCHU-PREDICATE + -ʔ T + TOPIC -TA NEW
pá-ʔchu-ʔ-ta =tsû jí-ya-mbi
die-EN-T-NEW =3 come-IRR-NEG
"If s/he's dead, s/he will not come." (`2024-06-20(2)_mll`)

5.6 *Adverbialized predicates*

Finally, I turn to adverbial constituents. Besides a closed class of lexical adverbs (37a), new A'ingae adverbs can be derived with the adverbializing suffix *-e* ADV (37b) or the adjectival adverbializer *-tshe* ADV.ADJ (37c). When an IS information structural morpheme attaches to an adverb, it is not preceded by a glottal stop *-ʔ* (37).

(37) PLAIN REALIZATION ON ADVERBS

a. ROOT ADVERB + NEW TOPIC -TA NEW
káni-nda =tsû án
yesterday-NEW =3 eat
"S/he ate yesterday." (`2025-07-22(1)_mll`)

14 The adnominal suffix *-a* ADN appears right after the nominalizer (in this case, *-ʔchu* EN) in nominalizations of predicates negated with *-mbi* NEG. For further discussion, see Dąbkowski (2023b).

b. *E*-ADVERB + CONTRASTIVE TOPIC -*JA* CNTR
tsínkun-ʔjen =tsû éga-e-ja
behave-IPFV =3 bad-ADV-CNTR
"S/he's behaving badly." (2025-07-22(1)_mll)

c. *TSHE*-ADVERB + EXCLUSIVE FOCUS -*YI* EXCL
gíya-tshe-yi =tsû áʔmbian-ña-ʔchu
clean-ADV.ADJ-EXCL =3 have-IRR-EN
"(One) must always keep (it) clean." (2025-07-22(1)_mll)

5.6.1 *Adverbialized nominalization*

A'ingae adverbial clauses can be divided into two broad classes: adverbialized nominalizations and negative adverbial clauses. Adverbialized nominalizations involve a clausal nominalization, often formed with the event nominalizer -*ʔchu* EN, followed by the adverbializing -*e* ADV (38). Adverbialized nominalizations can function as arguments of verbs such as *da* 'become' (38b) and *atesû* know (35g), same-subject rational clauses (38a, 38c), different-subject rational clauses (optionally introduced with *kûintsû/kaentsû* PURP.DS) (38d), hortatives introduced with *jinge(sû)* HORT (38e), and depictives (38f).

All of these uses are compatible with the discourse markers. When appearing on an adverbialized nominalization, regardless of its function, the new topic -*ta* NEW (38b), contrastive topic -*ja* CNTR (38c, 38e), and exclusive focus -*yi* EXCL (38d, 38f) are not preglottalized. The additive focus -*ʔkhe* ADD always has preglottalization (38g).

(38) PLAIN REALIZATION ON ADVERBIALIZED NOMINALIZATION WITH -*ʔCHU-E* EN-ADV

a. SAME-SUBJECT *ʔCHUE*-RATIONALE
áʔi=ja yusháva=ma kaváyu áyaʔfa=nga búta-en máni
person=CNTR iron=ACC horse mouth=DAT mouth-CAUS where
já-ya-ʔchu-ve tansi-án-ña-ʔchu-e
go-IRR-EN-ACC2 be straight-CAUS-IRR-EN-ADV
"When we put bits into the mouths of horses to make them obey us, we can turn the whole animal." (James 3:3; 2025-07-17(1)_mll)

b. *ʔCHUE*-ARGUMENT + NEW TOPIC -*TA* NEW
panzá-ya-ʔchu-e-ta =ngi dá
hunt-IRR-EN-ADV-NEW =1 become
"I was chosen to hunt." (2025-07-22(1)_mll)

c. SAME-SUBJECT *ʔCHUE*-RATIONALE + CONTRASTIVE TOPIC -*JA* CNTR
séma-ʔjen =ngi ánkheʔsû=ma áʔmbian-ña-ʔchu-e-ja
work-IPFV =1 food=ACC have-IRR-EN-ADV-CNTR
"I'm working to have food." (2025-08-21(1)_mll)

d. DIFFERENT-SUBJECT *ʔCHUE*-RATIONALE + EXCLUSIVE FOCUS *-YI* EXCL
séma-ʔjen =ngi (kû́intsû) tíse ánkheʔsû=ma áʔmbian-ña-ʔchu-e-yi
work-IPFV =1 PURP.DS 3SG food=ACC have-IRR-EN-ADV-EXCL
"I'm working so that s/he (can) have food." (2025-08-21(1)_mll)

e. *ʔCHUE*-HORTATIVE + CONTRASTIVE TOPIC *-JA* CNTR
jínge panzá-ya-ʔchu-e-ja
HORT hunt-IRR-EN-ADV-CNTR
"Let's hunt!" (2025-08-21(1)_mll)

f. *ʔCHUE*-DEPICTIVE + EXCLUSIVE FOCUS *-YI* EXCL
iyúfa-ʔkan-ʔchu-e-yi =tsû súmbu-ʔje
worm-SML-EN-ADV-EXCL =3 leave-IPFV
"It's coming out only like worms." (2025-08-12(1)_jxm)

g. *ʔCHUE*-ARGUMENT + ADDITIVE FOCUS *-ʔKHE* ADD
tíse =tsû panzá-ya-ʔchu-e-ʔkhe atésû
3SG =3 hunt-IRR-EN-ADV-ADD know
"S/he also knows how to hunt." (2025-08-21(1)_mll)

5.6.2 *Negative adverbial*

In addition to the adverbialized nominalizations discussed above, A'ingae also has negative adverbial clauses introduced with *-mbe* ADV.NEG (39). The negative adverbials are observed in two main contexts. First, they can function as adjuncts introducing "negative circumstance" clauses, which express an event that fails to accompany the event of the main clause (Fischer and Hengeveld, 2023) (39a, 39c, 39e, 39f). The negative circumstance clauses can often be translated as "not (doing sth)," "not having (done sth)," or "without (doing sth)." Second, the negative adverbials can be selected as arguments of the semantically bleached auxiliary verb *kan* AUX (39b, 39d). The meaning of clauses with *mbe*-arguments results from a composition of *-mbe*'s negation with the inflectional information present on *kan* AUX.

When the new topic *-ta* NEW (39b), contrastive topic *-ja* CNTR (39c), or exclusive focus *-yi* EXCL (39d, 39e) attaches to a negative adverbial clause (in either of its two functions), a glottal stop *-ʔ* is realized before the discourse marker.

(39) PREGLOTTALIZED REALIZATION ON NEGATIVE ADVERBIAL CLAUSES WITH *-MBE* ADV.NEG

a. NEGATIVE CIRCUMSTANCE *MBE*-ADJUNCT
simbá-ʔma án-mbi-si así-mbe jí
hook-FRST eat-NEG-DS fish-ADV.NEG come
"(I) was fishing with a hook, but (the fish) didn't take (the bait, so I) came (back home) without having caught (anything)."
(Borman and Chica Umenda, 1980, p. 42; 2025-07-22(1)_mll)

b. *MBE*-ARGUMENT + -ʔ T + NEW TOPIC *-TA* NEW
máʔkaen tsún-mba =ngi úshaʔchu pû̂iyiʔkhu=nga setsá-kheʔsû pákheʔsû=ma
how do-SS =1 everything everyone=DAT infect-HN disease=ACC
setsa-yé-mbe-ʔ-ta kán-ña?
infect-PASS-ADV.NEG-T-NEW AUX-IRR
"What can I do to avoid contracting contagious diseases?"
(Pederson and Cooper, 1982, p. 41; `2025-07-17(1)_mll`)

c. NEGATIVE CIRCUMSTANCE *MBE*-ADJUNCT + -ʔ T + CONTRASTIVE TOPIC *-JA* CNTR
kánse =tsû asitháen-ʔjen-mbe-ʔ-ja
live =3 worry-IPFV-ADV.NEG-T-CNTR
"S/he lives without a care in the world." (`2025-07-22(1)_mll`)

d. *MBE*-ARGUMENT + -ʔ T + EXCLUSIVE FOCUS *-YI* EXCL
sinthí-mbe-ʔ-yi tsû kán-ña-ʔchu kháse anjámpa tshán-saʔne
blow-ADV.NEG-T-EXCL =3 AUX-IRR-EN again blood bleed-APPR
"You should not blow your nose, so that you don't make it bleed again."
(Pederson and Cooper, 1982, pp. 68–69; `2024-02-20(1)_sia`)

e. NEGATIVE CIRCUMSTANCE *MBE*-ADJUNCT + -ʔ T + EXCLUSIVE FOCUS *-YI* EXCL
áthe-pa fíʔthi-mbe-ʔ-yi jí táyu kúse-ji-si
see-SS kill-ADV.NEG-T-EXCL come already night-INGR-DS
"I saw (the toucan but) I came (back home) without (having been able to) kill (it) because it was already getting dark."
(Borman and Chica Umenda, 1982a, p. 40; `2025-07-22(1)_mll`)

f. NEGATIVE CIRCUMSTANCE *MBE*-ADJUNCT + ADDITIVE FOCUS *-ʔKHE* ADD
kánse =tsû setsa-yé-mbe-ʔkhe
live =3 infect-PASS-ADV.NEG-ADD
"S/he does not get sick either." (`2025-07-17(1)_mll`)

Elsewhere, A'ingae negation is expressed with *-mbi* NEG, and adverbialization is achieved with *-e* ADV. Yet, the negative adverbial in the examples above is glossed as one morpheme (*-mbe* ADV.NEG), not as an adverbialization of a negative clause (**-mb-e* NEG-ADV) for two reasons. First, the negative *-mbi* NEG is compatible with the templatically preceding situation-level suffixes *-ʔfa* PLS (40a) and *-ya* IRR (40b). The negative adverbial *-mbe* ADV.NEG is not compatible with either (Hengeveld and Fischer, in prep.) (41) (also see Section 6.2). Morphotactic differences like this are unsurprising if *-mbe* ADV.NEG is a separate morpheme but would be unexplained if *-mbe* were a reduction of **-mb-e* NEG-ADV.

(40) NEGATIVE *-MBI* NEG COMPATIBLE WITH *-ʔFA* PLS, *-YA* IRR

a. *así-ʔfa-mbi*
fish-PLS-NEG
"(they) did not fish"
(`2025-11-06(1)_jxm`)

b. *así-ya-mbi*
fish-IRR-NEG
"will not fish"
(`2025-11-06(1)_jxm`)

(41) NEGATIVE ADVERBIAL -*MBE* ADV.NEG INCOMPATIBLE WITH -*ʔFA* PLS, -*YA* IRR

a. **así-ʔfa-mbe*
fish-PLS-ADV.NEG
intended: "without (them) fishing"
(2025-08-12(1)_jxm)

b. **así-ya-mbe*
fish-IRR-ADV.NEG
intended: "without being about to fish"
(2025-08-12(1)_jxm)

Second, a reduction of */-*mbi-e*/ to [-*mbe*] would be unexpected on phonological grounds. In A'ingae, the sequence [*ie*] forms a (rare but) licit diphthong. Notably, the sequence -*mbi-e* is actually attested when the adverbializing suffix -*e* ADV appears on *mbi*-final non-verbal constituents, such *tsaiʔmbi* 'many' (42a) and *khupa-mbi* 'defecate-NIN' "constipated" (42b). There are no independent reasons to believe that -*e* ADV should have a different phonological behavior on *mbi*-negated verbs than it does on *tsaiʔmbi* 'many' or *khupa-mbi* 'defecate-NIN.' Consequently, I conclude that -*mbe* ADV.NEG is a unitary morpheme.

(42) HETEROMORPHEMIC [*I-E*] LICIT IN ADVERBIALIZATION

a. LICIT [*I-E*] IN *TSAIʔMBI-E* 'MANY-ADV'
tsáiʔmbi-e bu-ñã́-mba khá-ne-ñi fíthi~ʔthi-ye ánshaen tetéte=ve=yi
many-ADV meet-CAUS-SS else-ELAT-EXCL kill~PLA-INF begin Waorani=ACC2=EXCL
"When he had piled up many of them, he began to kill the Waoranis from afar."
(Borman and Criollo, 1990, p. 173; 2025-07-22(1)_mll)

b. LICIT [*I-E*] IN *KHUPA-MBI-E* 'DEFECATE-NIN-ADV'
khupá-mbi-e =ti=ki dá?
defecate-NIN-ADV =YNQ=2 become
"Are you constipated?"
(Borman, Cooper, and Criollo, 1991, p. 37; 2025-08-12(1)_jxm)

In summary, in most contexts, the IS markers are not preceded by a glottal stop. The glottal stop appears only when a discourse marker attaches to an infinitival clause, a same-subject conditional antecedent (which lacks an overt subordinator), or a negative adverbial *mbe*-clause. These environments are repeated in (43).

(43) SUMMARY OF ENVIRONMENTS FOR IS MARKER PREGLOTTALIZATION

The IS markers are preglottalized only when attaching to:

a. infinitival clauses with -ye INF*,*

b. finite subordinate clauses which do not have an overt subordinator (i. e. same-subject conditional antecedents), and

c. negative adverbial clauses with -mbe ADV.NEG*.*

Otherwise, the IS markers are not preceded by a glottal stop.

6 ANALYSIS

In this section, I provide an analysis of the A'ingae IS-preglottalization pattern. In Section 6.1, I briefly discuss Bossi and Diercks's (2019) and Mikkelsen's (2015) accounts of information structure differentiation in in Kipsigis and Danish. In Section 6.2, I adopt their discourse feature geometry to capture the A'ingae data and present an analysis couched in Distributed Morphology (DM) (e. g. Embick and Noyer, 2007; Halle and Marantz, 1993, 1994). In Section 6.3, I present an application of the proposal to the A'ingae patterns.

6.1 *Discourse feature geometry*

In Danish (North Germanic, Denmark), in some contexts, the clause-initial position may be filled by different phrases, including e. g. discourse categories such as topic and focus, expressions related to illocutionary force (such as WH-words), and constituents that establish a rhetorical relation to the preceding sentence (such as adverbs) (Mikkelsen, 2015, p. 21), but not expletives or discourse-old subjects (Mikkelsen, 2015, p. 12). For example, in the continuation of (44), the fronted constituent can be the VP anaphor *det* VPA (44a) or a rhetorical adverb such as *heldigvis* 'luckily' (44b). However, the expletive *der* 'there' cannot be fronted (44c). In the examples below, the fronted constituents are underlined.

(44) FRONTING OF DISCOURSE-DISTINGUISHED CONSTITUENTS

Danish (North Germanic, Denmark)

da jeg åbnede døren troede jeg først at der havde [*været indbrud*]$_{VP}$, *men* ...
when 1SG opened door.DEF thought 1SG first that there had been break in but
"When I opened the door, I first thought that someone had broken into the house but ... "

a. LICIT FRONTING OF THE VP ANAPHOR *DET* VPA
<u>det</u> havde der heldigvis ikke
VPA had there luckily not

b. LICIT FRONTING OF A RHETORICAL ADVERB
<u>heldigvis</u> havde der ikke det
luckily had there not VPA

c. ILLICIT FRONTING OF A DISCOURSE-UNDISTINGUISHED EXPLETIVE
**<u>der</u> havde* (*det*) *heldigvis ikke* (*det*)
there had VPA luckily not VPA
"luckily there hadn't (been a break-in)." (Mikkelsen, 2015, p. 19)

To capture the above data, Mikkelsen (2015) proposes that "[in] Danish, Spec,CP must be occupied by an information-structurally distinguished element, but is not dedicated to a particular function" (p. 634). In other words, the notion relevant for capturing which constituents can surface in the clause-initial position is that of general underspecified *information structural differentiation*, which can be satisfied in a variety of ways (e. g. by the aforementioned information-structural, illocutionary, or rhetorical properties).

Bossi and Diercks (2019) observe an empirically similar pattern in Kipsigis (Nilo-Saharan, Kenya), where a discourse-prominent item is likewise fronted. In the verb-initial Kipsigis, however, the landing site for the fronted constituent is the immediately postverbal position (IPP). As in Danish, the fronted distinguished constituent can have one of many discourse functions. For example, when answering a WH-question (46), the IPP has to be occupied by the new information focus (46a). A non-focus constituent in the IPP is infelicitous (46b).

(45) IMMEDIATELY POSTVERBAL NEW INFORMATION FOCUS Kipsigis (Nilo-Saharan, Kenya)
kii-∅-gootʃi ŋoo kibeet kɪtabʊʊt?
PST-3SG-give who Kibeet book
"Who gave Kibeet a book?"

a. NEW INFORMATION FOCUS FRONTED
kii-∅-gootʃi tʃeepkoetʃ kibeet kɪtabʊʊt
PST-3SG-give Chepkoech Kibeet book
"Chepkoech gave Kibeet a book."

b. ILLICIT FRONTING OF NON-FOCUS
#*koo-∅-gootʃi kɪtabʊʊt tʃeepkoetʃ kibeet*
PST-3SG-give book Chepkoech Kibeet
intended: "Chepkoech gave Kibeet a book." (Bossi and Diercks, 2019, p. 8)

However, when an aboutness topic is forced (e. g. by saying "tell something about ... ") (46), the IPP is preferentially occupied by the topic (46a). Using the dispreferred word order of (46b) "generates a sense that the speaker is abruptly forcing a shift in topic" (Bossi and Diercks, 2019, p. 11).

(46) IMMEDIATELY POSTVERBAL ABOUTNESS TOPIC Kipsigis (Nilo-Saharan, Kenya)
mwɔ-ɔn kiit agɔbɔ kiproono
tell-1SG.OBJ thing about Kiproono
"Tell me something about Kiproono."

a. ABOUTNESS TOPIC FRONTED
kii-∅-min kiproono baandeek
PST-3SG-plant Kiproono maize
"Kiproono planted the maize."

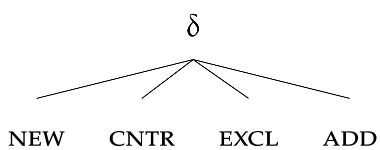
b. ILLICIT FRONTING OF NON-TOPIC
#?*kii-∅-min baandeek kiproono*
PST-3SG-plant maize Kiproono
intended: "Kiproono planted the maize." (Bossi and Diercks, 2019, p. 11)

To account for the discourse-driven word order of Kipsigis, Bossi and Diercks (2019) build on Mikkelsen's (2015) insight and formalize *information structural differentiation* as "an underspecified [δ] (discourse) feature that can be satisfied by phrases of any information structure designation or—even more generally—by any phrase that is sufficiently discourse-differentiated" (p. 30).[15] Concretely, the movement to the IPP is modeled as movement to Spec,TP triggered by an uninterpretable [δ] borne by T° (Bossi and Diercks, 2019, p. 30).

6.2 *Distributed Morphology formalization*

In my analysis of the A'ingae patterns, I adopt Bossi and Diercks's (2019) feature geometry directly and propose that the four IS morphemes (*-ta* NEW, *-ja* CNTR, *-yi* EXCL, *-ʔkhe* ADD) are the exponents of four discourse features: [NEW], [CNTR], [EXCL], and [ADD]. The IS features are organized hierarchically, with a superordinate *discourse* feature [δ] dominating the four more specific features. A tree representing a minimally differentiated organization of the relevant IS features is repeated in (47).[16] As in all feature-geometric systems, I assume that the presence of a lower feature entails the presence of the hierarchically higher one.

(47) DISCOURSE FEATURE GEOMETRY



I propose that the discourse feature [δ] is the context in which T° is realized as a glottal stop *-ʔ*. I assume a decompositional approach to T, where overt situation-level suffixes are realized as heads below T° (48).[17]

Specifically, I assume that every clause—by definition—contains the TP layer (Shlonsky, 1997, p. 3). Additionally, I assume that TP dominates the projections which realize negation (Laka Mugarza, 1990; Pollock, 1989) and irrealis mood (Cinque, 1999), and that the hierarchical order of the number (#P), mood (MP), and polarity (ΣP) projections mirrors the linear order of suffixes they introduce (*-ʔfa* PLS, *-ya* IRR, and *-mbi* NEG). Taken together, the above premises zero in on the finite TP structure given in (48a).

The A'ingae infinitive *ye*-clauses can be marked for subject plurality (*-ʔfa* PLS), but not for reality (**-ya* IRR) or negation (**-mbi* NEG). Following Wurmbrand (1998), I assume that TP dominates the projection that realizes finiteness. Infinitive TP structure is given in (48b).

15 A similar claim is made by Kim (2015) for the Korean marker -(*n*)*un*. While -(*n*)*un* is traditionally analyzed as a contrastive topic, Kim (2015) proposes that -(*n*)*un* only serves to "impose salience on a discourse referent" and that "[t]opicality and contrast [are] … derived from the interaction of the proposed meaning of -(*n*)*un* and various syntactic, semantic, and pragmatic factors" (p. 87).

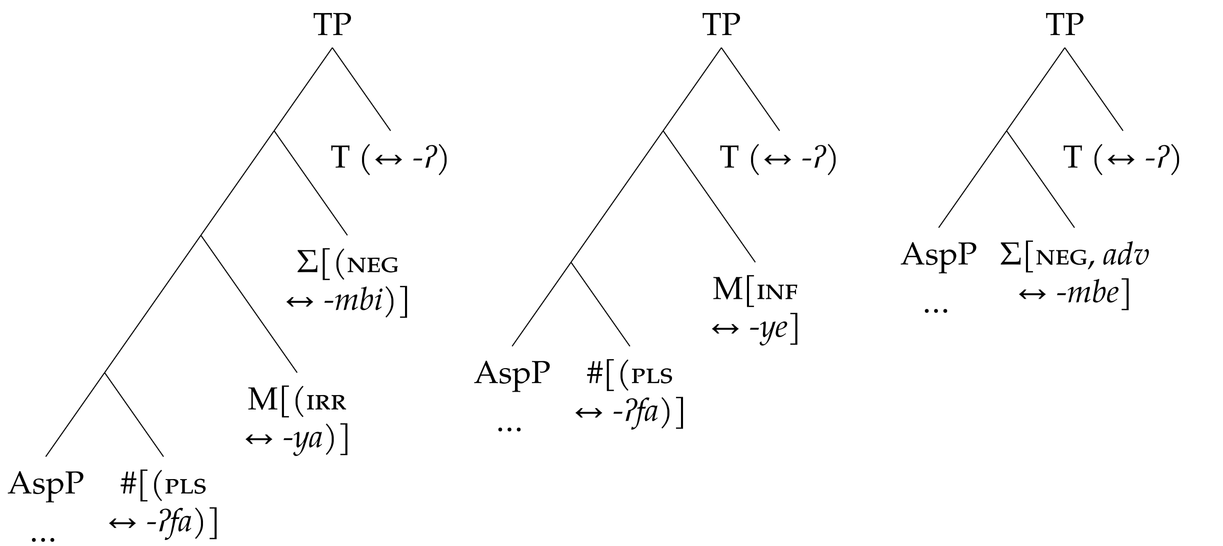
16 Although the figure in (47) shows minimally differentiated discourse feature organization, the proposal is compatible with more hierarchical systems. For example, following Krifka (2008), one may take contrastive topic [CNTR] to inherit from higher-order topic [TOP] and focus [FOC] features.

17 While T° is often silent, its presence correlates in A'ingae with specific temporal interpretations. For example, while uninflected stative and non-verbal TPs are interpreted as present, uninflected eventive verbs are interpreted as realis, perfective, and past. For more, see Dąbkowski (in prep.).

The negative adverbial *mbe*-clauses cannot be marked for subject plurality (*-*ʔfa* PLS) or reality (*-*ya* IRR). Since the negative adverbial marker *-mbe* ADV.NEG is a single morpheme (see §5.6.2), I assume that it is a realization of one syntactic head that hosts both the negative [NEG] and adverbial categorizing [*adv*] features. The structure of the negative adverbial TPs is given in (48c).

(48) TP STRUCTURE

a. FINITE TPs b. INFINITIVE TPs c. NEGATIVE ADVERBIAL TPs



As previously stated, I propose that the glottal stop *-ʔ* discussed throughout this paper is a contextual realization of T°. Concretely, T° is realized as a glottal stop *-ʔ* when linearly adjacent to the discourse feature [δ]. The critical vocabulary item (VI) is given in (49a). Since *-ta* NEW, *-ja* CNTR, *-yi* EXCL, and *-ʔkhe* ADD all inherit from [δ], they all satisfy this environment condition. Otherwise, T° is realized as phonologically null (49b). This derives the observed distribution of the IS-conditioned *-ʔ*. All VIs relevant to the analysis are given in (49-51).[18]

(49) FEATURES ⪯ TP

a. T ⟷ *-ʔ* / _ δ
b. T ⟷ *-∅* / elsw.
c. PLS ⟷ *-ʔfa*
d. IRR ⟷ *-ya*
e. INF ⟷ *-ye*
f. NEG, *adv* ⟷ *-mbe*
g. NEG ⟷ *-mbi*

(50) C-HEADS

a. IF, SS ⟷ *-ʔa* / _ ADD
b. IF, SS ⟷ *-∅* / elsw.
c. SS ⟷ *-pa*
d. IF, DS ⟷ *-ʔni*
e. DS ⟷ *-si*
f. APPR ⟷ *-saʔne*
g. TENT ⟷ *-khen*

(51) DISCOURSE FEATURES

a. NEW ⟷ *-ta*
b. CNTR ⟷ *-ja*
c. EXCL ⟷ *-yi*
d. ADD ⟷ *-ʔkhe*

18 One could alternatively consider an analysis where the topic and focus features inherit from a general Ā-feature, and *-ʔ* is the realization of T° when linearly adjacent to [Ā] (i. e. T ⟷ *-ʔ* / _ Ā). For a short discussion of this alternative, see Section A.2.

6.3 *Application of the proposal*

Now, I will demonstrate how the proposed analysis accounts for the data. When an IS marker attaches to most syntactic categories, including DPs (52a) and adverbs (52b), there is no T° adjacent to a discourse-marked morpheme. As such, *-ʔ* is not realized. The constituent to which the IS markers are attached is bracketed [ ].

(52) MOST CATEGORIES: *-ʔ* NOT REALIZED

a. DP + NEW TOPIC *-TA* NEW

[yáya]$_{DP}$=ta =tsû tsámpi=ni já
dad=NEW =3 forest=LOC go
"Dad went hunting." (2024-06-10(1)_mll)

b. ADVERB + CONTRASTIVE TOPIC *-JA* CNTR

[tayúpi]$_{advP}$-ja fíthi~ʔthi =tsû
long ago-CNTR kill~PLA =3
"Once upon a time, s/he killed (many)." (2024-10-08(1)_mll)

When an IS marker attaches immediately to an infinitive TP (e. g. a complement of a raising predicate, such as the habitual auxiliary *atesû* 'know') (53a), the adjacent T-head is realized as *-ʔ* (49a). In the examples below, spans stretching from T° through the discourse markers are underlined.

(53) RAISING PREDICATE COMPLEMENT: *-ʔ* REALIZED

a. *-ʔ* T + NEW TOPIC *-TA* NEW

[ñúʔfa-ye-ʔ]$_{TP}$-ta =ngi atésû
rest-INF-T-NEW =1 know
"I (habitually) rest." (2024-10-08(1)_mll)

When an IS marker attaches to CP with an overt subordinating C°, there is overt material intervening between T° and the IS marker (54a). Since the environments conditioning allomorphy are strictly local (Embick, 2010, 2015), *-ʔ* is not realized (49b). (The assumption that infinitival morphology is TP-internal, while subordinating morphemes are C-heads, is widely accepted cross-linguistically (Adger, 2003), and corroborated for A'ingae by Dąbkowski, 2022, 2024b.)

(54) OVERTLY-SUBORDINATED CLAUSE: *-ʔ* NOT REALIZED

a. NULL-T + OVERT C + EXCLUSIVE FOCUS *-YI* EXCL

[já-ya-∅-pa]$_{CP}$-yi =ngi ína-ʔjen
go-IRR-T-SS-EXCL =1 cry-IPFV
"I'm crying only because I will leave." (2024-10-08(1)_mll)

However, when an IS marker attaches to a CP with a null C°, such as a complement of a control predicate, e. g. *seʔpi* 'forbid' (55a), there is no overt morphology intervening between T° and the IS marker. I assume that phonologically unrealized material is ignored for purposes of satisfying allomorphy environments (*pruning* in Embick, 2010, 2015). As such, T° is realized as *-ʔ*.[19]

(55) CONTROL PREDICATE COMPLEMENT: *-ʔ* REALIZED

a. *-ʔ* T + NULL-C + CONTRASTIVE TOPIC *-JA* CNTR

[*kéʔi án-ʔfa-ye-ʔ-∅*]$_{CP}$*-ja* *séʔpi* *=ngi*
2PL eat-PL-INF-T-C-CNTR forbid =1
"I prohibit y'all from eating." (2024-10-08(1)_mll)

Next, I propose that same-subject conditional antecedents are likewise introduced by a head that is phonologically unrealized in most environments (50b). Since, again, null material is ignored for the purposes of allomorphy (Embick, 2010, 2015), T° is realized as *-ʔ* (56a-56b).

(56) SAME-SUBJECT CONDITIONAL ANTECEDENT: *-ʔ* REALIZED

a. *-ʔ* T + NULL-C + CONTRASTIVE TOPIC *-JA CNTR*

[*afa-khú-ʔ-∅*]$_{CP}$*-ja* *sumbú-ya =ngi*
speak-RCPR-T-IF.SS-CNTR leave-IRR =1
"If I argue, I will leave." (2024-06-11(2)_mll)

b. *-ʔ* T + NULL-C + EXCLUSIVE FOCUS *-YI EXCL*

[*thési-ma* *áthe-ʔ-∅*]$_{CP}$*-yi* *=ngi bûthú-ya*
jaguar-ACC see-T-IF.SS-EXCL =1 run-IRR
"As soon as I see a jaguar, I will run." (2024-10-08(1)_mll)

This derives the fact that *-ʔ* T is realized on same-subject antecedents, but not on other types of subordinate clauses—only the same-subject antecedents are introduced by a phonologically unrealized feature bundle.[20]

---

19 Since silent heads are ignored for allomorph selection, the status of any clause as a TP or a silent-C° CP is immaterial for the analysis. Nevertheless, throughout the section, I represent phonologically null C-heads in contexts where they would be assumed by standard analyses of phenomena such as control, clausal adjunction, etc. (e. g. Adger, 2003).

20 Cross-linguistically, conditionals and topics are marked in similar, often identical, ways (Haiman, 1978). Under this analysis, A'ingae conditionality is encoded with dedicated morphosyntactic features (50a-b, 50d), different from the features responsible for discourse-related notions (51). Nevertheless, the IS markers are obligatorily on same-subject antecedents (§5.3.5) and optional (though frequent) on different-subject antecedents (§5.3.6).
The present analysis does not formally capture the fact that discourse marking on same-subject antecedents is obligatory. However, since the feature bundle expressing same-subject antecedent semantics is in most contexts phonologically unrealized, i. e. [IF, SS] ⟷ -∅ (50b), I speculate that the obligatory IS marking on same-subject antecedents may be functionally motivated: If there were no discourse markers, the same-subject antecedents would be morphologically indistinguishable from matrix clauses.

When several discourse markers are attached to a TP, the glottal stop *-ʔ* is realized only before the first one (57a). This is a direct consequence of the fact that *-ʔ* is a realization of T° (and there is normally only one T° per clause).

(57) MULTIPLE IS MARKERS: *-ʔ* REALIZED ONLY ONCE

a. *-ʔ* T + NULL-C + EXCLUSIVE FOCUS *-YI EXCL* + NEW TOPIC *-TA* NEW

[*thési=ma áfase-ʔ-∅*]$_{CP}$*-yi-ja bûthú-ya =ngi*

jaguar=ACC criticize-T-IF.SS-EXCL-CNTR run-IRR =1

"Only if I criticize a jaguar, I will run." (`2024-06-06(2)_mll`)

When a same-subject antecedent hosts the additive focus *-ʔkhe* ADD, the morpheme *-ʔa* appears before it (58a). I propose that *-ʔa* is an allomorph of the same-subject conditional feature bundle introduced specifically in the context of *-ʔkhe* ADD (50a).

(58) SAME-SUBJECT CONDITIONAL ANTECEDENT REALIZED AS *-ʔA* BEFORE *-ʔKHE* ADD

a. NULL-T + *-ʔA* IF.SS + ADDITIVE FOCUS *-ʔKHE* ADD

[*khúpa-ʔnga-pa jí-∅-ʔa*]$_{CP}$*-ʔkhe =tsû utishí-ya-ʔchu*

defecate-AND-SS come-T-IF.SS-ADD =3 wash hands-IRR-EN

"After using the restroom, one must also wash hands."

(Borman and Chica Umenda, 1982b, p. 27; `2024-06-07(1)_mll`)

When an IS marker attaches to a non-final verb in an SVC, no glottal stop is realized between the verb and the marker (59a). Non-final serialized verbs can host VceP and AspP morphology, but not TP or CP morphology, which reveals that they are structurally small AspPs. As such, they have no T° that could be realized as *-ʔ*.

(59) SVC-INTERNAL VERB: *-ʔ* NOT REALIZED

a. NON-FINAL SERIALIZED VERB + NEW TOPIC *-TA* NEW

[*fû́ndu-ʔje*]$_{AspP}$*-ta jí*

sceam-IPFV-NEW come

"(S/he) came screaming." (`2025-08-28(1)_mll`)

When an information structural marker attaches to a nominalized TP, there is a nominalizer, e. g. the bounded space *-khû* BND (60a), intervening between T° and the marker. As such, T° is not realized as a glottal stop.

(60) NOMINALIZED CLAUSE: *-ʔ* NOT REALIZED

a. NULL-T + NOMINALIZER + NEW TOPIC *-TA* NEW

[*án-ña-∅-khû*]$_{DP}$*-ta =tsû ránde*

eat-IRR-T-BND-NEW =3 large

"The room where (s/he) will eat is large." (`2025-08-12(1)_jxm`)

When a nominalizer intervenes between the T-head and a discourse marker, T° is not realized as a glottal stop (61a). However, the nominalized clause can serve as a complement to another T° when it functions as a predicate (e. g. in a same-subject antecedent). The second T° is realized as a glottal stop *-ʔ* if linearly adjacent to a discourse marker (61b). Thus, the difference between the members of a minimal pair, such as (61), follows directly from their structural differences.

(61) MINIMAL *ʔCHU*(*ʔ*)*TA*-PAIR

a. DP + NEW TOPIC *-TA* NEW
[pá-∅-ʔchu]$_{DP}$-ta =tsû jí-ya-mbi
die-T-EN-NEW =3 come-IRR-NEG
"(The) dead (one) will not come." (2024-06-20(2)_mll)

b. DP + *-ʔ* T + NULL-C + NEW TOPIC *-TA* NEW
[pá-∅-ʔchu-ʔ-∅]$_{CP}$-ta =tsû jí-ya-mbi
die-T-EN-T-IF.SS-NEW =3 come-IRR-NEG
"If s/he's dead, s/he will not come." (2024-06-20(2)_mll)

When a discourse marker attaches to an adverbialized nominalization, there are two morphemes intervening between the T-head and the discourse marker (62a). As such, a glottal stop is not realized.

(62) ADVERBIALIZED NOMINALIZATION: *-ʔ* NOT REALIZED

a. NULL-T + *-ʔCHU* EN + *-E* ADV + CONTRASTIVE TOPIC *-JA* CNTR
[panzá-ya-∅-ʔchu-e]$_{advP}$-ta =ngi dá
hunt-IRR-T-EN-ADV-NEW =1 become
"I was chosen to hunt." (2025-07-22(1)_mll)

I assume that the negative adverbial *-mbe* ADV.NEG is a single exponent which is introduced in the polarity projection (ΣP) and expresses both the negative [NEG] and adverbial categorizing [*adv*] features. Since ΣP is below TP, when a discourse marker attaches to a negative adverbial clause, no overt material intervenes between the T-head and the discourse marker. As such, T° is realized as a glottal stop *-ʔ* (63a).

(63) NEGATIVE ADVERBIAL: *-ʔ* REALIZED

a. *-MBE* ADV.NEG + *-ʔ* T + NULL-C + CONTRASTIVE TOPIC *-JA* CNTR
kánse =tsû [asitháen-ʔjen-mbe-ʔ-∅]$_{CP}$-ja
live =3 worry-IPFV-ADV.NEG-T-C-CNTR
"S/he lives without a care in the world." (2025-07-22(1)_mll)

Finally, I propose that *-ʔkhe* ADD is underlyingly preglottalized (51d). I assume that *-ʔkhe* ADD participates in the same triggering of T-allomorphy as the other IS morphemes (64a), but two contiguous glottal stops are later phonologically reduced to one.

(64) ADDITIVE FOCUS -*ʔKHE* ADD: UNDERLYINGLY PREGLOTTALIZED

a. -*ʔ* T + NULL-C + ADDITIVE FOCUS -*ʔKHE* ADD

[*án-ñe-ʔ-∅*]$_{CP}$*-ʔkhe sé̱ʔpi =ngi*

eat-INF-T-C-ADD forbid =1

"I also forbid eating." (2024-06-11(2)_mll)

In summary, we have seen that the glottal stop -*ʔ* appears at the right edge of TP when immediately followed by an IS morpheme. To capture this distribution, I have proposed that the glottal stop -*ʔ* is a realization of the T-head conditioned by linear adjacency to the discourse feature [δ]. The schema representing these structural conditions is repeated in (65).

(65) *ʔ*-REALIZATION SCHEMA

right edge of TP ↘

[*root* ... **-*ʔ***]$_{TP}$ -*sfx*$_{[\delta]}$

↑ ↑ ↖ information structural morpheme

T° no overt material intervening

7 DISCUSSION AND CONCLUSIONS

In this paper, I have described and analyzed the conditions under which a glottal stop -*ʔ* appears before IS morphemes. Specifically, I have shown that the glottal stop is realized only if a discourse morpheme is attached directly to a TP, and can be analyzed as the spell-out of T° when linearly adjacent to the superordinate discourse feature [δ]. The proposed analysis has several theoretical implications.

First, it draws on the notion of *discourse differentiation*, proposed by Mikkelsen (2015) and formalized as [δ] by Bossi and Diercks (2019). By showing that all of the A'ingae IS morphemes pattern alike in triggering allomorphy of T°, the paper provides novel morphological evidence for a geometric arrangement of discourse features, with [δ] inherited by all the maximal ones. In doing so, the present study contributes to the broader research program on the geometric arrangement of morphosyntactic features (e. g. Béjar and Rezac, 2009; Harley and Ritter, 2002; Preminger, 2011), and of Ā-features in particular (e. g. Aravind, 2018; Baclawski, 2019; Baier, 2018).

Second, the contextual realization of -*ʔ* provides support for less restrictive theories of allomorph selection. In Section 6, the glottal stop -*ʔ* has been shown to be a realization of T° when immediately followed by a discourse morpheme. Under this analysis, the allomorphy of T° is conditioned by an outer morpheme (i. e. a morpheme attaching *after* T°). This is in direct violation of the Peripherality Constraint (Carstairs, 1987, p. 196; Kiparsky, 2021, p. 392), which states allomorphy can be triggered only inside-out. The Peripherality Constraint is quoted in (66).

(66) PERIPHERALITY CONSTRAINT (Kiparsky, 2021, p. 392 from Carstairs, 1987, p. 193)

"The realization of a property P may be sensitive inwards, i. e. to a property realized more

centrally in the word-form (that is, closer in linear sequence to the root) but not outwards to an individual property realized more peripherally (further from the root)."

The proposed analysis is, however, compatible with less restrictive views of allomorphy that allow for outward-looking selection. Outward-looking allomorph selection has been shown to be necessary for a number of languages, including Itelmen (Chukotko-Kamchatkan; Bobaljik, 2000), Luganda (Bantu; Hyman, 2015), Nez Perce (Sahaptian; Deal and Wolf, 2017), and others.[21]

Third, by referring to independently motivated morphosyntactic features (such as [T] and [δ]) and utilizing widely accepted mechanisms (such as *pruning*; Embick, 2010, 2015), a DM analysis can model the somewhat complex distribution of the A'ingae *-ʔ* with a single vocabulary item. This reduction of an empirically novel pattern to a combination of analytical tools needed to account for unrelated phenomena in unrelated languages bears out the predictions of syntax-centric models of morphology in a novel way.

At the same time, the A'ingae *-ʔ* T remains a descriptive and typological rarum. Its distribution is abstract and difficult to state in non-theoretical terms. Using relatively descriptive language, the glottal stop *-ʔ* could be said to mark the morphosyntactic boundary between "predicate inflection" and discourse-level marking. However, without invoking verb-internal syntactic structure, the notion of "predicate inflection" is difficult to state precisely,[22] and the requirement on "discourse-level marking" can be satisfied by one of several semantically related suffixes. In other words, the A'ingae pattern highlights the value of highly abstract work in revealing the distribution of otherwise puzzling morphology.

Finally, the present proposal makes a non-obvious descriptive claim related to conditional morphology: Even though same-subject antecedents are seen most often with the preglottalized *-ʔta* and *-ʔja*, neither sequence is analyzed as contributing the conditional meaning (67a).[23] Rather, the glottal stop *-ʔ* is proposed to be a syntactically conditioned spell-out of

---

21 If, on the other hand, the analysis were flipped on its head, and the glottal stop *-ʔ* analyzed as a spell-out of [δ] in the presence of a linearly adjacent (and preceding) T-head, i. e. [δ] ⟷ *-ʔ* / T _, the Peripherality Constraint (66) would be satisfied. However, since T is clearly a syntactic (and syntactically relevant) feature, the revised analysis would run afoul of Bobaljik's (2000, pp. 2–3) second generalization (ib).

(i) Bobaljik's (2000, pp. 2–3) generalizations
  a. *"[A]llomorphy conditioned by more peripheral morphemes/features is only sensitive to morpho-**syntactic** (i. e., syntactically relevant) features (such as tense and agreement)[;]*
  b. *"inwards-sensitive allomorphy ... is conditioned by morpho-(**phono-**)logical features such as class marking and other (syntactically irrelevant) diacritics[.]"* (emphasis Bobaljik's)

Additionally, note that since the IS morphemes all inherit from [δ], the suffixes *-ta*, *-ja*, *-yi*, *-ʔkhe* already spell out [δ]. As such, adopting this alternative—where *-ʔ* is also a realization of [δ]—would involve multiple exponence.

22 For example, on one hand, no particular overt verbal morpheme is required for the glottal stop *-ʔ* to be realized. On the other hand, the glottal stop is not realized on non-terminal verbs in SVCs and or on verbs overtly (co)subordinated with morphemes other than the infinitive *-ye* INF. Yet, notably, this disjunctive, messy list of conditions can easily be dispensed with by simply saying that the glottal stop *-ʔ* is a spell-out of T°.

the T-head, the discourse marker is taken to contribute its regular discourse meaning, and the same-subject conditional morpheme is taken to actually be phonologically silent (67b).

(67) ANALYSIS INFORMS GLOSSING

a. INCORRECT GLOSSING

afa-khú-ʔja *sumbú-ya =ngi*
*speak-RCPR-IF.SS leave-IRR =1
"If I argue, I will leave." (2024-06-11(2)_mll)

b. CORRECT GLOSSING

afa-khú-ʔ-∅-ja *sumbú-ya =ngi*
speak-RCPR-T-IF.SS-CNTR leave-IRR =1
"If I argue, I will leave." (2024-06-11(2)_mll)

This reiterates the important—though perhaps uncontroversial (e. g. Bach, 2004; Dryer, 2006; Hanson, 1979; Kuhn, 1962; Mendikoetxea, 2003)—point that language description and theoretical analysis are not two separate endeavors, and that they must mutually inform each other.

BIBLIOGRAPHY

Adger, David (2003). *Core syntax: A Minimalist approach*. Oxford Core Linguistics. Oxford University Press.

Aikhenvald, Alexandra Y. (Oct. 2018). *Serial Verbs*. Oxford University Press. ISBN: 9780198791263. DOI: 10.1093/oso/9780198791263.001.0001. URL: https://doi.org/10.1093/oso/9780198791263.001.0001.

Aissen, Judith (2023). "Documenting topic and focus." In: *Key topics in language documentation and description*. Ed. by Peter Jenks and Lev Michael. Language Documentation & Conservation Special Publication 26 [PP 11–57]. University of Hawai'i Press: Honolulu. URL: https://hdl.handle.net/10125/75032.

AnderBois, Scott, Daniel Altshuler, and Wilson de Lima Silva (2023). "The forms and functions of switch reference in A'ingae." In: *Languages* 8.2. ISSN: 2226-471X. DOI: 10.3390/languages8020137. URL: https://www.mdpi.com/2226-471X/8/2/137.

AnderBois, Scott and Maksymilian Dąbkowski (2020). "A'ingae =*sa'ne* APPR and the semantic typology of apprehensional adjuncts." In: *Proceedings of the 30th Semantics and Linguistic Theory Conference*. Ed. by Joseph Rhyne, Kaelyn Lamp, Nicole Dreier, and Chloe Kwon. Washington, DC: Linguistic Society of America, pp. 43–62. DOI: 10.3765/salt.v30i0.4804.

23 In this, the present analysis differs from some previous proposals (e. g. Dąbkowski, 2021a,b, 2024b), which make the mistake of presenting *-ʔta* and *-ʔja* as morphologically atomic.

AnderBois, Scott and Maksymilian Dąbkowski (2025). "The semantics and expression of apprehensional modality." In: *Language and Linguistics Compass* 19.1, e70002. DOI: https://doi.org/10.1111/lnc3.70002. eprint: https://compass.onlinelibrary.wiley.com/doi/pdf/10.1111/lnc3.70002. URL: https://compass.onlinelibrary.wiley.com/doi/abs/10.1111/lnc3.70002.

AnderBois, Scott, Nicholas Emlen, Hugo Lucitante, Chelsea Sanker, and Wilson de Lima Silva (2019). "Investigating linguistic links between the Amazon and the Andes: The case of A'ingae." Paper presented at the 9th Conference on Indigenous Languages of Latin America. University of Texas at Austin.

AnderBois, Scott and Wilson de Lima Silva (2018). "Kofán Collaborative Project: Collection of audio-video materials and texts." London: SOAS, Endangered Languages Archive. URL: https://elar.soas.ac.uk/Collection/MPI1079687.

Aravind, Athulya (2018). "Licensing long-distance *wh*-in-situ in Malayalam." In: *Natural Language & Linguistic Theory* 36.1, pp. 1–43. DOI: 10.1007/s11049-017-9371-2. URL: http://hdl.handle.net/1721.1/113876.

Asher, Nicholas and Alex Lascarides (2003). *Logics of Conversation*. Cambridge University Press. ISBN: 9780521659512. URL: https://philpapers.org/rec/ASHLOC.

Bach, Emmon (2004). "Linguistic universals and particulars." In: *Linguistics Today – Facing a Greater Challenge*. Ed. by Piet van Sterkenburg. John Benjamins Publishing Company, pp. 47–60. ISBN: 9789027295149. DOI: doi:10.1075/z.126.04bac. URL: https://doi.org/10.1075/z.126.04bac.

Baclawski Jr., Kenneth Paul (2019). "Discourse connectedness: The syntax–discourse structure interface." PhD thesis. University of California, Berkeley. URL: https://escholarship.org/uc/item/4hx630c2.

Baier, Nicholas B. (2018). "Anti-agreement." PhD thesis. University of California, Berkeley.

Béjar, Susana and Milan Rezac (2009). "Cyclic Agree." In: *Linguistic Inquiry* 40.1, pp. 35–73.

Bennett, Ryan, Shen Aguinda, Hugo Lucitante, and Scott AnderBois (2025). "Phonetic evidence for contextual underspecification of nasality in A'ingae." Manuscript.

Blaser, Magdalena and María Enma Chica Umenda (2008). *A'indeccu canqque'sune condase'cho. Mitos del pueblo cofán*. Centro Cultural de Investigaciones Indígenas Padre Ramón López.

Bobaljik, Jonathan David (2000). "The ins and outs of contextual allomorphy." In: *University of Maryland Working Papers in Linguistics* 10, pp. 35–71.

Borman, Marlytte "Bubs" and Enrique Criollo (1990). *La cosmología y la percepción histórica de los cofanes de acuerdo a sus leyendas*. Cuadernos etnolingüísticos 10. Quito, Ecuador: Instituto Lingüístico de Verano (Summer Institute of Linguistics). URL: https://www.sil.org/resources/archives/17586.

Borman, Roberta "Bobbie" (1981). *Aprendamos (Cofán – Castellano)*. Quito, Ecuador: Republica del Ecuador Ministerio de Educación y Cultura (Republic of Ecuador Department of Education and Culture) and Instituto Lingüístico de Verano (Summer Institute of Linguistics). URL: https://www.sil.org/resources/archives/17831.

Borman, Roberta "Bobbie" and María Enma Chica Umenda (1977). *A'ingae I: Cofán. Texto de lectura I: Cofán – castellano*. Quito, Ecuador: Ministerio de Educación Pública. URL: https://www.sil.org/resources/archives/17822.

Borman, Roberta "Bobbie" and María Enma Chica Umenda (1980). *A'ingae III: Cofán. Texto de lectura III: Cofán – castellano*. Quito, Ecuador: Ministerio de Educación y Cultura. URL: https://www.sil.org/resources/archives/17754.

Borman, Roberta "Bobbie" and María Enma Chica Umenda (1982a). *A'ingae II: Cofán. Texto de lectura II: Cofán – castellano*. Quito, Ecuador: Ministerio de Educación y Cultura. URL: https://www.sil.org/resources/archives/50932.

Borman, Roberta "Bobbie" and María Enma Chica Umenda (1982b). *A'ingae V: Cofán. Texto de lectura V: Cofán – castellano*. Quito, Ecuador: Ministerio de Educación y Cultura. URL: https://www.sil.org/resources/archives/51279.

Borman, Roberta "Bobbie", Verla Cooper, and Emeregildo Criollo (1991). *El cofán y el médico: Vocabulario médico bilingüe cofán–castellano*. Quito, Ecuador: Instituto Lingüístico de Verano (Summer Institute of Linguistics). URL: https://www.sil.org/resources/archives/11117.

Bossi, Madeline and Michael Diercks (2019). "V1 in Kipsigis: Head movement and discourse-based scrambling." In: *Glossa: A Journal of General Linguistics* 4.1. DOI: 10.5334/gjgl.246.

Carstairs, Andrew (1987). *Allomorphy in Inflexion*. Croom Helm Linguistics Series. London: Croom Helm, pp. xv + 271. ISBN: 9780415825108.

Chica Umenda, María Enma (1980). *Chimbi a'inga attian'cho toya'caen fuesu condase'cho* [*El murciélago y la mujer y otras leyendas cofanes*]. Cofán 4. Quito, Ecuador: Instituto Lingüístico de Verano (Summer Institute of Linguistics). URL: https://www.sil.org/resources/archives/17710.

Cinque, Guglielmo (1999). *Adverbs and functional heads: A cross-linguistic perspective*. Oxford Studies in Comparative Syntax. Oxford University Press. ISBN: 9780195354058. URL: https://books.google.com/books?id=sFd79vhd_n0C.

Criollo, Emergildo and Claudino Blanco Pisabarro (1992). *Ingi ta gi a'indeccu'fa. Nosotros los cofanes*. Quito, Ecuador: Federación Indígena de la Nacionalidad Cofán del Ecuador.

Crippen, James A. (2019). "The syntax in Tlingit verbs." PhD thesis. Vancouver: University of British Columbia.

Dąbkowski, Maksymilian (2020). "A'ingae field materials." 2020–19. California Language Archive, Survey of California and Other Indian Languages. University of California, Berkeley. DOI: 10.7297/X2HH6HKG.

Dąbkowski, Maksymilian (2021a). "Conditional constructions in A'ingae." Manuscript. University of California, Berkeley. URL: https://ling.auf.net/lingbuzz/006349.

Dąbkowski, Maksymilian (2021b). "Dominance is non-representational: Evidence from A'ingae verbal stress." In: *Phonology* 38.4, pp. 611–650. DOI: 10.1017/S0952675721000348.

Dąbkowski, Maksymilian (2022). "A'ingae pied-piping: A Q-based analysis." Paper presented at the 4th Symposium on Amazonian Languages. University of California, Berkeley. URL: https://lingbuzz.net/lingbuzz/008650.

Dąbkowski, Maksymilian (2023a). "A'ingae reduplication is phonologically optimizing." In: *Supplemental Proceedings of the 2022 Annual Meeting on Phonology*. Ed. by Noah Elkins, Bruce Hayes, Jinyoung Jo, and Jian-Leat Siah. Washington, DC: Linguistic Society of America. DOI: 10.3765/amp.v10i0.5450.

Dąbkowski, Maksymilian (2023b). "Polarity agreement in A'ingae nominalizations." Paper presented at the Annual Meeting of the Society for the Study of the Indigenous Languages of the Americas. Denver, CO. URL: https://tinyurl.com/dabkowski2023ssila.

Dąbkowski, Maksymilian (2023c). "Postlabial raising and paradigmatic leveling in A'ingae: A diachronic study from the field." In: *Proceedings of the Linguistic Society of America*. Ed. by Patrick Farrell. Vol. 8. 1. 5428. Washington, DC: Linguistic Society of America. DOI: 10.3765/plsa.v8i1.5428.

Dąbkowski, Maksymilian (2024a). "The phonology of A'ingae." In: *Language and Linguistics Compass* 18.3, e12512. ISSN: 1749-818X. DOI: 10.1111/lnc3.12512. eprint: https://compass.onlinelibrary.wiley.com/doi/pdf/10.1111/lnc3.12512. URL: https://compass.onlinelibrary.wiley.com/doi/abs/10.1111/lnc3.12512.

Dąbkowski, Maksymilian (2024b). "Two grammars of A'ingae glottalization: A case for Cophonologies by Phase." In: *Natural Language and Linguistic Theory* 42.2, pp. 437–491. ISSN: 1573-0859. DOI: 10.1007/s11049-023-09574-5.

Dąbkowski, Maksymilian (2025a). "A Q-Theoretic solution to A'ingae postlabial rounding." In: *Linguistic Inquiry*, pp. 1–14. ISSN: 0024-3892. DOI: 10.1162/ling_a_00550. eprint: https://direct.mit.edu/ling/article-pdf/doi/10.1162/ling_a_00550/2482649/ling_a_00550.pdf. URL: https://doi.org/10.1162/ling_a_00550.

Dąbkowski, Maksymilian (2025b). "Deglottalizing contamination in A'ingae historical derivatives." In: *Proceedings of the Linguistic Society of America*. Vol. 10. 1. Washington, DC: Linguistic Society of America, p. 5879. DOI: 10.3765/plsa.v10i1.5879. URL: https://journals.linguisticsociety.org/proceedings/index.php/PLSA/article/view/5879.

Dąbkowski, Maksymilian (2025c). "Metrical stress and glottal stops in A'ingae: A study of cyclicity and dominance at the interface of phonology and morphology." PhD thesis. University of California, Berkeley. URL: https://escholarship.org/uc/item/4w34v92f.

Dąbkowski, Maksymilian (2025d). "Morphological boundary glottals in A'ingae: A new argument for [δ]." In: *NELS 55: Proceedings of the Fifty-Fifth Annual Meeting of the North Eastern Linguistic Society, Volume One*. Ed. by Duygu Demiray, Roger Cheng-yen Liu, and Nir Segal. Amherst, Massachusetts: GLSA (Graduate Linguistics Student Association), Department of Linguistics, University of Massachusetts, Amherst, pp. 149–158. URL: https://ling.auf.net/lingbuzz/009006.

Dąbkowski, Maksymilian (2025e). "Phasal strength in A'ingae classifying subordination." In: *Proceedings of the 2023 and 2024 Annual Meetings on Phonology*. Ed. by Gérard Avelino, Merlin Balihaxi, Quartz Colvin, Vincent Czarnecki, Hyunjung Joo, Chenli Wang, Utku Zobarlar, Adam Jardine, and Adam McCollum. Vol. 1. 1. Amherst, Massachusetts: University of Massachusetts Amherst Libraries. DOI: 10.7275/amphonology.3012. URL: https://openpublishing.library.umass.edu/amphonology/article/id/3012/.

Dąbkowski, Maksymilian (in prep.). "A'ingae classifiers and the limits of dominance: A first look at syntactically blocked morphophonology." In: *Phonological domains in Algonquian and other morphologically complex languages*. Ed. by Natalie Weber. Oxford University Press. URL: https://lingbuzz.net/lingbuzz/008770.

Dąbkowski, Maksymilian (t.a.). "A'ingae second-position clitics are matrix C-heads." In: *Proceedings of the 25th Workshop on Structure and Constituency in the Languages of the Americas*. Vancouver, BC: University of British Columbia Working Papers in Linguistics. URL: https://ling.auf.net/lingbuzz/006463.

Dąbkowski, Maksymilian and Scott AnderBois (2024). "Rationale and precautioning clauses: Insights from A'ingae." In: *Journal of Semantics* 40.2-3, pp. 391–425. ISSN: 0167-5133. DOI: 10.1093/jos/ffac012. eprint: https://academic.oup.com/jos/advance-article-pdf/doi/10.1093/jos/ffac012/56485636/ffac012.pdf. URL: https://doi.org/10.1093/jos/ffac012.

Dąbkowski, Maksymilian and Scott AnderBois (t.a.). "The apprehensional domain in A'ingae (Cofán)." In: *Apprehensional constructions in a cross-linguistic perspective*. Ed. by Martina Faller, Marine Vuillermet, and Eva Schultze-Berndt. Research on Comparative Grammar. Berlin: Language Science Press, pp. 77–118. URL: https://ling.auf.net/lingbuzz/006450.

Deal, Amy Rose and Matthew Wolf (2017). "Outwards-sensitive phonologically-conditioned allomorphy in Nez Perce." In: *The Morphosyntax-Phonology Connection: Locality and Directionality at the Interface*. Ed. by Vera Gribanova and Stephanie S. Shih. Oxford University Press, pp. 29–60.

Dryer, Matthew S. (2006). "Descriptive theories, explanatory theories, and Basic Linguistic Theory." In: *Catching Language: The Standing Challenge of Grammar Writing*. Ed. by Felix K. Ameka, Alan Dench, and Nicholas Evans. Vol. 167. Trends in Linguistics. Berlin, New York: De Gruyter Mouton, pp. 207–234. ISBN: 9783110197693. DOI: doi:10.1515/9783110197693.207. URL: https://doi.org/10.1515/9783110197693.207.

Embick, David (2010). *Localism versus Globalism in Morphology and Phonology*. Linguistic Inquiry Monograph 60. Cambridge, MA: MIT Press. URL: https://mitpress.mit.edu/9780262514309/localism-versus-globalism-in-morphology-and-phonology.

Embick, David (2015). *The Morpheme: A Theoretical Introduction*. Interface Explorations [IE] 31. Boston and Berlin: De Gruyter Mouton. DOI: 10.1515/9781501502569.

Embick, David and Rolf Noyer (2007). "Distributed morphology and the syntax-morphology interface." In: *The Oxford Handbook of Linguistic Interfaces*, pp. 289–324.

Fischer, Rafael (2007). "Clause linkage in Cofán (A'ingae), a language of the Ecuadorian-Colombian border region." In: *Indigenous Languages of Latin America (ILLA)*. Vol. 5: *Language Endangerment and Endangered Languages: Linguistic and anthropological studies with special emphasis on the languages and cultures of the Andean-Amazonian border area*. Ed. by W. Leo Wetzels. Leiden: CNWS Publications, pp. 381–399. URL: https://hdl.handle.net/11245/1.277948.

Fischer, Rafael and Kees Hengeveld (2023). "A'ingae (Cofán/Kofán)." In: *Amazonian Languages: An International Handbook*. Vol. 1: *Language Isolates I: Aikanã to Kandozi-Shapra*.

Ed. by Patience Epps and Lev Michael. Handbooks of Linguistics and Communication Science (HSK) 44. Berlin: De Gruyter Mouton, pp. 65–124. ISBN: 9783110419405. DOI: 10.1515/9783110419405.

Fischer, Rafael and Eva van Lier (2011). "Cofán subordinate clauses in a typology of subordination." In: *Subordination in Native South American Languages*. Ed. by Rik van Gijn, Katharina Haude, and Pieter Muysken. Typological Studies in Language 97. Amsterdam/Philadelphia: John Benjamins, pp. 221–250. DOI: 10.1075/tsl.97.09fis.

Haiman, John (1978). "Conditionals are topics." In: *Language* 54.3, pp. 564–589. DOI: 10.2307/412787.

Halle, Morris and Alec Marantz (1993). "Distributed Morphology and the pieces of inflection." In: *The View from Building 20: Essays in Linguistics in Honor of Sylvain Bromberger*. Ed. by Kenneth Hale and Samuel Jay Keyser. Cambridge, Massachusetts: MIT Press, pp. 111–176.

Halle, Morris and Alec Marantz (1994). "Some key features of Distributed Morphology." In: *MIT Working Papers in Linguistics* 21.275, p. 88.

Hanson, Norwood Russell (1979). *Patterns of Discovery: An Inquiry into the Conceptual Foundations of Science*. Cambridge University Press.

Harley, Heidi and Elizabeth Ritter (2002). "Person and number in pronouns: A feature-geometric analysis." In: *Language* 78.3, pp. 482–526.

Hengeveld, Kees and Rafael Fischer (2018). "A'ingae (Cofán/Kofán) operators." In: *Open Linguistics* 4.1, pp. 328–355. URL: https://www.degruyter.com/view/journals/opli/4/1/article-p328.xml.

Hengeveld, Kees and Rafael Fischer (in prep.). *A grammar of A'ingae*. University of Amsterdam.

Hyman, Larry M. (2015). "Outward-looking *y*/∅ alternations in Luganda." Talk presented at the 46th Annual Conference on African Linguistics. University of Oregon.

Jasinskaja, Katja and Elena Karagjosova (2020). "Rhetorical relations." In: *The Wiley Blackwell Companion to Semantics*. John Wiley & Sons, Ltd, pp. 1–29. ISBN: 9781118788516. DOI: https://doi.org/10.1002/9781118788516.sem061. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781118788516.sem061. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118788516.sem061.

Kalpande, Arushi (2024). "The semantics of A'ingae serial verb constructions." Honors thesis. Providence, RI: Brown University.

Karsten, Nils (2021). "Accusative marking – Nominal mood in A'ingae." In: *Linguistics in Amsterdam* 14.2, pp. 1–27. URL: https://linguisticsinamsterdam.nl/home?issue=142.

Kehler, Andrew (2001). *Coherence, Reference, and The Theory of Grammar*. Stanford, CA: CSLI Publications, p. 231. ISBN: 1-57586-216-6.

Kim, Ilkyu (2015). "Is Korean -(*n*)*un* a topic marker? On the nature of -(*n*)*un* and its relation to information structure." In: *Lingua* 154, pp. 87–109. ISSN: 0024-3841. DOI: https://doi.org/10.1016/j.lingua.2014.11.010. URL: https://www.sciencedirect.com/science/article/pii/S0024384114002769.

Kiparsky, Paul (2021). "Phonology to the rescue: Nez Perce morphology revisited." In: *The Linguistic Review* 38.3, pp. 391–442.

Krifka, Manfred (2008). "Basic notions of information structure." In: *Acta Linguistica Hungarica* 55.3-4, pp. 243–276.

Kuhn, Thomas S. (1962). *The Structure of Scientific Revolutions*. Chicago: University of Chicago Press, p. 264. ISBN: 9780226458113.

Laka Mugarza, Miren Itziar (1990). "Negation in syntax: On the nature of functional categories and projections." PhD thesis. Massachusetts Institute of Technology. URL: https://dspace.mit.edu/bitstream/handle/1721.1/13667/24302516-MIT.pdf?sequence=2.

Lovestrand, Joseph (2021). "Serial verb constructions." In: *Annual Review of Linguistics* 7, pp. 109–130.

Lucitante, Bolivar et al. (2011). *La Gente del Río: La historia del Sararo y la Choni contada por el pueblo Cofán* (*Naensu Ai: Tayopisu cuenzandeccu sararo chonine condasecho*). Quito, Ecuador: Pontificia Universidad Católica del Ecuador. ISBN: 978-9978-77-176-1.

Mendikoetxea, Amaya (2003). "On the intricate relation between theory and description: A linguist's look at The Cambridge Grammar of the English Language." In: *Círculo de Lingüística Aplicada a la Comunicación* 16, pp. 3–34. ISSN: 1576-4737. URL: https://revistas.ucm.es/index.php/CLAC.

Mikkelsen, Line (2015). "VP anaphora and verb-second order in Danish." In: *Journal of Linguistics* 51.3, pp. 595–643. DOI: 10.1017/S0022226715000055.

Morvillo, Sabrina and Scott AnderBois (2022). "The inner workings of contrast: Decomposing A'ingae *tsa'ma*." In: *Proceedings of Semantics of Understudied Languages of the Americas (SULA) 11*, pp. 171–186. URL: https://research.clps.brown.edu/anderbois/PDFs/MorvilloAnderBois.pdf.

Pederson, Lois and Verla Cooper (1982). "Quinsetsse canseye toya seje'suni jambi'te tsoñe." Quito, Ecuador: Instituto Lingüístico de Verano (Summer Institute of Linguistics). URL: https://www.sil.org/resources/archives/42661.

Pollock, Jean-Yves (1989). "Verb movement, universal grammar, and the structure of IP." In: *Linguistic inquiry* 20.3, pp. 365–424. ISSN: 00243892, 15309150. URL: http://www.jstor.org/stable/4178634.

Preminger, Omer (2011). "Agreement as a fallible operation." PhD thesis. Cambridge, MA: Massachusetts Institute of Technology. URL: http://hdl.handle.net/1721.1/68196.

Quenamá, Gregorio (1992). *Gregorio Quenamá: Mi historia*. Ed. by Delfín Criollo and Roberta "Bobbie" Borman. Autobiografías Cofanes 2. Quito, Ecuador: Publicaciones Cofanes. URL: https://www.sil.org/resources/archives/41791.

Repetti-Ludlow, Chiara, Haoru Zhang, Hugo Lucitante, Scott AnderBois, and Chelsea Sanker (2019). "A'ingae (Cofán)." In: *Journal of the International Phonetic Association: Illustrations of the IPA*, pp. 1–14. DOI: 10.1017/S0025100319000082.

Repetti-Ludlow, Chiara (2021). "The A'ingae co-occurrence constraint." In: *Supplemental Proceedings of the 2020 Annual Meeting on Phonology*. Ed. by Ryan Bennett, Richard Bibbs, Mykel L. Brinkerhoff, Max J. Kaplan, Stephanie Rich, Amanda Rysling, Nicholas Van

Handel, and Maya Wax Cavallaro. Vol. 8. Washington, DC: Linguistic Society of America. DOI: 10.3765/amp.v9i0.4859.

Rivet, Paul (1924). *Langues américaines*. Vol. 2: *Langues de l'Amérique du Sud et des Antilles*. Ed. by Antoine Meillet and Marcel Cohen. Les langues du monde. Société Linguistique de Paris.

Rivet, Paul (1952). "Affinités du Kofán." In: *Anthropos* 47 (1/2), pp. 203–234.

Ruhlen, Merritt (1987). *A Guide to the World's Languages; Vol. 1: Classification*. Stanford, California: Stanford University Press. ISBN: 9780804718943. URL: https://books.google.com/books?id=mYwmDE3f6wUC.

Sanker, Chelsea and Scott AnderBois (2024). "Reconstruction of nasality and other aspects of A'ingae phonology." In: *Cadernos de Etnolingüística* 11, e110101. URL: http://www.etnolinguistica.org/article:vol11n1.

Shlonsky, Ur (1997). *Clause structure and word order in Hebrew and Arabic: An essay in comparative Semitic syntax*. Oxford University Press.

The Bible (1980). Chiga Tevaen'jen. Las Sagradas Escrituras en el idioma Cofán del Ecuador. Wycliffe Inc.

van Kuppevelt, Jan (1995). "Discourse structure, topicality and questioning." In: *Journal of Linguistics* 31.1, pp. 109–147. DOI: 10.1017/S002222670000058X.

von Stutterheim, Christiane and Wolfgang Klein (1989). "Referential movement in descriptive and narrative discourse." In: *Language Processing in Social Context*. Ed. by Rainer Dietrich and Carl F. Graumann. Vol. 54. North-Holland Linguistic Series: Linguistic Variations. Amsterdam: North Holland, pp. 39–76. ISBN: 9781483295374. DOI: 10.1016/B978-0-444-87144-2.50005-7. URL: https://www.sciencedirect.com/science/article/pii/B9780444871442500057.

Wurmbrand, Susanne (1998). "Infinitives." PhD thesis. Massachusetts Institute of Technology. URL: https://dspace.mit.edu/bitstream/handle/1721.1/9592/42141033-MIT.pdf?sequence=2&isAllowed=y.

Zheng, Holly and Scott AnderBois (2023). "Definiteness in A'ingae and its implications for pragmatic competition." In: *Formal Approaches to Languages of South America*. Ed. by Cilene Rodrigues and Andrés Saab. Cham: Springer International Publishing, pp. 347–371. ISBN: 978-3-031-22344-0. DOI: 10.1007/978-3-031-22344-0_13. URL: https://doi.org/10.1007/978-3-031-22344-0_13.

ACKNOWLEDGMENTS I would like to wholeheartedly thank members of the Cofán communities whose generosity and assistance have made this research uniquely possible. Thanks in particular to Jorge Criollo and his family for welcoming me to their home, my primary collaborators on this project—Marcelo Lucitante and Shen Aguinda—for their insight, patience, and kindness, as well as Hugo Lucitante for support with all matters along the way.

I would also like to thank Peter Jenks, Line Mikkelsen, Scott AnderBois, Kees Hengeveld, Hannah Sande, Carol Rose Little, and the audiences at the 2025 Annual Meeting of the Society for the Study of the Indigenous Languages of the Americas (SSILA 2025), the 55th Annual Meeting of the North East Linguistic Society (NELS 55), the Berkeley Syntax and

Semantics Circle (SSCircle), the A'ingae Language Documentation Project (ALDP) lab, and the Stanford Syntax and Morphology Circle (SMircle) for their invaluable feedback and helpful discussions.

FUNDING This project was supported in part by National Science Foundation 20-538 Linguistics Program's Doctoral Dissertation Research Improvement grant #2314344 to Maksymilian Dąbkowski for *Doctoral Dissertation Research: Nominal and deverbal morphology in an endangered language*, American Philosophical Society's Lewis and Clark Fund to Maksymilian Dąbkowski for Exploration and Field Research on *Stress and glottalization across lexical classes in A'ingae (or Cofán; Ecuador)*, and California Language Archive's Oswalt Endangered Language Grants to Maksymilian Dąbkowski for fieldwork research on the *Form and productivity in A'ingae derivation* and *A'ingae nominalizers: Their present and their past*.

A APPENDIX

A.1 *Uses of discourse markers*

A'ingae has four morphemes dedicated to encoding discourse-related information: two encoding topichood (whose basic uses are sketched in Section A.1.1), and two—focus (described in Section A.1.2).

A.1.1 *Topic markers*

The topic of a sentence is the entity that the sentence is intuitively "about." This is to say, the topic is a referent that the rest of the sentence comments on (Krifka, 2008). A'ingae distinguishes two types of overtly marked topics: the new topic *-ta* NEW and the contrastive topic *-ja* CNTR. The semantic contributions of the markers are often not reflected in translations to Spanish or English. Moreover, in most sentences, *-ta* NEW and *-ja* CNTR can be removed without resulting in ungrammaticality. Below, I discuss some prototypical uses of the two topic markers.

The new topic marker *-ta* NEW indicates that the referent was not previously present in the discourse. For example, the marker often appears on individuals that are introduced for the first time (68).

(68) REFERENT INTRODUCED WITH *-TA* NEW
[seeing a city in the distance for the first time:]
jú-va *t́útu-a=<u>ta</u>* *=ti* *patû́-ndûʔndû*
DIST-DEM white-ADN=NEW =YNQ rock-beach
"Is that white over there a rocky beach?"
(Borman and Chica Umenda, 1982b, p. 24; `2025-08-26(1)_mll`)

The above observation is supported by native speakers' metalinguistic judgments. For example, when topichood is forced with a simple "what about" question, answers with the new topic *-ta* NEW (69b) are preferred over information structurally unmarked (69a) and contrastive topic *ja*-marked answers (69c).[24]

(69) ANSWER TO "WHAT ABOUT." *TA*-MARKING PREFERRED
[A: What about Edison?]

a. DISCOURSE-UNMARKED RESPONSE
#B: *Édison =tsû yúku cháthû=ʔsû já*
Edison =3 yoco cut=SN go

b. NEW TOPIC-MARKED RESPONSE
B: *Édison=nda =tsû yúku cháthû=ʔsû já*
Edison=NEW =3 yoco cut=SN go

c. CONTRASTIVE TOPIC-MARKED RESPONSE
#B: *Édison=jan yúku cháthû=ʔsû =tsû já*
Edison=CNTR yoco cut=SN =3 go

"Edison left to cut some yoco." (`2025-08-22(1)_mll`)

The new topic marker *-ta* NEW is also commonly observed on "stage-setting" constituents, which specify the time or location of an event. In such contexts, speakers explicitly judge *ta*-marked phrases (70b) as better than the unmarked (70a) and *ja*-marked ones (70c).

(70) STAGE SETTING: *TA*-MARKING PREFERRED
[beginning of a chapter:]

a. DISCOURSE-UNMARKED GROUND
#tayúpi =tsû injántshi Saráru kánse-ʔfa Avássicu náʔen=ni
long ago =3 many giant otter live-PLS Aguarico river=LOC

b. NEW TOPIC-MARKED GROUND
tayúpi-ta =tsû injántshi Saráru kánse-ʔfa Avássicu náʔen=ni
long ago-NEW =3 many giant otter live-PLS Aguarico river=LOC

c. CONTRASTIVE TOPIC-MARKED GROUND
#tayúpi-ja injántshi Saráru =tsû kánse-ʔfa Avássicu náʔen=ni
long ago-CNTR many giant otter =3 live-PLS Aguarico river=LOC

"In the past, many otters lived in the Aguarico River."
(based on Lucitante et al., 2011, pp. 48–49; `2025-08-22(1)_mll`)

The contrastive topic marker *-ja* CNTR is used in the presence of alternative topics in the discourse, especially when information about different individuals is overtly or covertly juxtaposed. For example, it may be used on a subject which contrasts with other discourse

24 A constituent marked with the contrastive topic *-ja* CNTR cannot be directly followed by a second position clitic. As a consequence, for example, in (69c), the clitic is shifted down to the following constituent. This is further discussed in Section 3.

participants in whether they took part in some activity (71). In comparable circumstances, English may distinguish the contrasted constituent prosodically.

(71) SUBJECT PROPERTY CONTRASTED WITH -JA CNTR
[dialogue between A and B:
A: Who all went?
B: Dad, my uncle, and my older brother.]
A: *míngaʔu-si =ki ké=ja já-mbi*
why-DS =2 2SG=CNTR go-NEG
"Why didn't *you* go?"
(Borman and Chica Umenda, 1982b, p. 9; `2025-08-26(1)_mll`)

Another use of the contrastive topic *-ja* CNTR can be seen in (72). The paragraph's main question under discussion (van Kuppevelt, 1995; von Stutterheim and Klein, 1989) is: *what are tapirs like?* The subquestions about different body part of the tapir, such as *what are tapirs' noses like?* (72a) and *what are tapirs' feet like?* (72b), receive answers with different body parts explicitly marked with the contrastive topic *-ja* CNTR.

(72) SUBQUESTIONS CONTRASTED WITH -JA CNTR
[description of a tapir: "Tapirs live in the jungle. They frequent salt licks ...]
a. *tsá=ta =tsû tíse tsúfaʔthû=ja biá-ʔa ...*
ANA=NEW =3 3SG nose=CNTR long-ADN
"Their noses are long ... "
b. *túyaʔkaen tíse tsû́ʔthe=ja táʔe-tshi*
and 3SG foot=CNTR hard-ADJ
"And their feet are hard."
(Borman and Chica Umenda, 1982b, p. 33; `2025-08-26(1)_mll`)

These patterns are, again, confirmed by speakers' metalinguistic judgements. When different individuals are contrasted with one another, *ja*-marking (73c) is preferred over *ta*-marking (73b) or lack of marking (73a). Below, this is illustrated with a question-and-answer pair, where the answer explicitly juxtaposes two people.

(73) JUXTAPOSED PROPERTIES: JA-MARKING PREFERRED
[A: Do your children work hard?]
a. DISCOURSE-UNMARKED TOPICS
#B: *ña únkeʔnge =tsû pánsheʔen semá-ñe atésû; ña dû́tshiʔye =tsû nuéʔsû*
1SG daughter =3 a lot work-INF know 1SG son =3 be lazy
b. TA-MARKED TOPICS
#B: *ña únkeʔnge=ta =tsû pánsheʔen semá-ñe atésû; ña dû́tshiʔye=ta =tsû nuéʔsû*
1SG daughter=NEW =3 a lot work-INF know 1SG son=NEW =3 be lazy

c. JA-MARKED TOPICS
B: *ña únkeʔnge=ja pánsheʔen =tsû semá-ñe atésû; ña dûtshiʔye=ja nuéʔsû =tsû*
1SG daughter=CNTR a lot =3 work-INF know 1SG son=CNTR be lazy =3
"My daughter works a lot; my son is lazy." (2025-08-26(1)_mll)

Similarly, in narrative contexts, when attention is drawn to an opposition between two (or more) entities, *ja*-marking on the contrasted participants is judged as more felicitous than a lack of marking or *ta*-marking (74b, 74c).

(74) CONTRASTED PARTICIPANTS: JA-MARKING PREFERRED
[beginning of the story *Peccary Demons:* "Four hunters went deep into the forest in search of game, and they lost their way. As it was about to get dark, two of them started to set up camp, and the other two left to look for food ...]

a. *tsá-ʔka-en já-pa shishithúshe=ma áthe-pa*
ANA-SML-ADV go-SS chameleon=ACC see-SS
"(The two hunters who were looking for food) came upon a chameleon ... "

b. *fáeʔkhu{#=∅, #=ta, =ja} índi-pa fíthi~ʔthi*
one=∅, =NEW, =CNTR catch-SS kill~PLA
"One (of them) grabbed (and started) killing (it) ... "

c. *tsá=ʔma fáesû{#=∅, #=ta, =ja} phuráen-mbi*
ANA=FRST other=∅, =NEW, =CNTR touch-NEG
"but (the) other (one) didn't touch (it)."
(based on Blaser and Chica Umenda, 2008, p. 123; 2025-08-26(1)_mll)

The felicity conditions on the discourse unmarked, *ta*-marked, and *ja*-marked constituents can be further illustrated by seeing how the specifics of a situation may affect the way a speaker chooses to introduce themselves. When asked to introduce themselves out of the blue (with no additional context provided), an information structurally unmarked response is felicitous (75a). A *ta*-marked first-person subject can be used in an introduction at the beginning of a conference talk (i. e. when one is shifting to a new topic) (75b). And lastly, in a group setting after hearing out other people's introductions, one may opt to set oneself apart with the contrastive *ja*-marking (75c).

(75) INTRODUCTIONS

a. UNMARKED INTRODUCTION
[A: Who are you?]
B: *Marcélo =ngi*
Marcelo =1
"I'm Marcelo." (2025-08-22(1)_mll)

b. *TA*-MARKED INTRODUCTION
[Speaker is at a conference. He is about to give a talk about his work in the Indigenous Cofán Guard in front of a large crowd. He goes on state and introduces himself:]
ñá=nda =ngi Marcélo
1SG=NEW =1 Marcelo
"I'm Marcelo." (2025-08-22(1)_mll)

c. *JA*-MARKED INTRODUCTION
[Speaker and others sit in a circle at a meeting. Everyone in the group introduces themself. First, one person says "I'm Maksymilian," another one says "I'm Edison," another one "I'm Leidy." Then, the speaker introduces himself:]
ñá=jan Marcélo =ngi
1SG=CNTR Marcelo =1
"I'm Marcelo." (2025-08-22(1)_mll)

Finally, I note that *ta-* and *ja*-marked constituents are infelicitous as fragment answers to WH-questions (76). This is consistent with their topic status (which is incompatible with the focus information of fragment answers; Aissen, 2023).

(76) NO TOPIC MARKING IN FRAGMENT ANSWERS
[A: Who came?]
B: *ña ántian{=∅, #=nda, #=jan}*
1SG brother=∅, =NEW, =CNTR
"My brother." (2025-08-21(1)_mll)

A.1.2 *Focus markers*

Now, I briefly discuss the A'ingae focus markers. Focus indicates that in addition to the referent denoted by the marked phrase, other entities are also relevant to the interpretation of the sentence (Krifka, 2008, p. 247). A'ingae has two focus morphemes: the exclusive focus *-yi* EXCL and the additive focus *-ʔkhe* ADD. The exclusive focus *-yi* EXCL indicates that a proposition holds of the marked entity to the exclusion of alternatives; it can often be translated as "only" (77a), "just" (77b), "very (same)" (77c), "as soon as" (77d), or with cleft constructions (77e).

(77) USES OF THE EXCLUSIVE FOCUS *-YI* EXCL
a. *tsá-ʔka-en pá-ʔfa-si khuánifae tsandié=ndekhû=yi khûshá-ʔfa*
ANA-SML-ADV die-PLS-DS three man=PL.ANIM=EXCL survive-PLS
"When they thus died, only three men were spared."
(Blaser and Chica Umenda, 2008, p. 37; 2024-06-07(1)_mll)

b. *túya án-ʔjen-mba-yi =ki jí-mbi?*
still eat-IPFV-SS-EXCL =2 come-NEG
"You didn't come just because you were still eating?" (2024-06-10(1)_mll)

c. *tsá-ʔkan-si =tsû típe-ʔsû áʔi=yi=ja máʔkaen fíʔthi-pa ávû kíniʔjin=ma júkhaningae tsún-ña-ʔchu-ve-ja tsún-ʔfa*
ANA-SML-DS =3 3SG-ATTR person=EXCL=CNTR somehow kill-SS fish tree=ACC for later do-IRR-EN-ACC2-CNTR do-PLS
"Because of that, those very same ones had to kill, ensuring the survival of the fish tree." (Blaser and Chica Umenda, 2008, p. 25; 2024-06-07(1)_mll)

d. *panzá-ʔ-yi-ta =ngi buthû́-ya*
hunt-T-EXCL-NEW =1 run-IRR
"As soon as I catch it, I will run." (2025-08-12(1)_jxm)

e. *junguésû=yi =tsû náʔen=ni kánse?*
what=EXCL =3 river=LOC live
"What is it that lives in the river?" (2024-06-10(1)_mll)

The additive focus *-ʔkhe* ADD indicates that the proposition holds of the marked entity in addition to some alternatives; it can often be translated as "too," "as well," "also" (78a), "and" (78b), or "even (though)" (78c), and—in negative polarity contexts—as "either" (78d) or "(not) even" (78e). (The additive focus *-ʔkhe* ADD is analyzed as underlyingly preglottalized, so the overt presence or absence of *-ʔ* T can never be observed before it. See Section 6 for the full analysis.)

(78) Uses of the additive focus *-ʔkhe* ADD

a. *tsá=ma áthe-pa fáesû=ma=ʔkhe áthe*
ANA=ACC see-SS other=ACC=ADD see
"When (he) looked, (he) also saw another one."
(Blaser and Chica Umenda, 2008, p. 38; 2024-06-07(1)_mll)

b. *tíse =tsû kínse-tshi túyaʔkaen tsúve ínjan-ʔjen-ʔchu-ʔkhe ñú-tshi*
3SG =3 healthy-ADJ and head like-IPFV-EN-ADD good-ADJ
"S/he is healthy and mentally well." (2025-07-22(1)_mll)

c. *kúndase khûi, tíse dyú-ʔni-ʔkhe*
tell lie down 3SG fear-IF.DS-ADD
"While lying (on the ground, the severed head) talked (to him/her), even though s/he was scared." (Chica Umenda, 1980, p. 26; 2024-03-04(1)_sia)

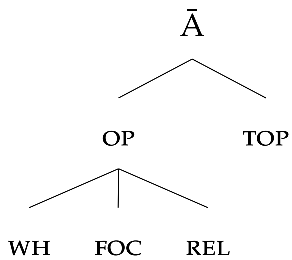
d. *kánse =tsû setsa-yé-mbe-ʔkhe*
live =3 infect-PASS-ADV.NEG-ADD
"S/he does not get sick either." (2025-07-17(1)_mll)

e. *tsá-ʔma tsé-ki kúse ni fae ávû=ve=yi=ʔkhe índi-ʔfa-mbi =ngi*
ANA-FRST STA-DRN night neither one fish=ACC2=EXCL=ADD catch-PLS-NEG =1
"But that night we did not catch even a single fish."
(John 21:3; `2024-06-07(1)_mll`)

A.2 *Ā-geometry analysis*

Topic and focus also figure in other feature geometries, such as (79), where [Ā] dominates [TOP], [FOC] (and hence presumably also [NEW], [CNTR], [EXCL], and [ADD]), as well as other features. As such, one might ask if *-ʔ* could alternatively be analyzed as the realization of T° when linearly adjacent to [Ā] (i. e. T ⟷ *-ʔ* / _ Ā).

(79) Ā-FEATURE GEOMETRY (Baier, 2018, p. 45, adapted from Aravind, 2018, p. 19)



This alternative analysis predicts that T° should be realized as *-ʔ* when adjacent to suffixes expressing not only topic and focus, but also other Ā-features, such as [WH] or [REL].

This prediction is not easy to test because overt morphemes which express Ā-features—other than the information structural *-ta* NEW, *-ja* CNTR, *-yi* EXCL, and *-ʔkhe* ADD—(almost) never appear to the immediate right of a verb-internal T°. For example, WH-bearing constituents are realized as independent morphosyntactic words, and the language lacks overt relativizers altogether.

One morphosyntactic configuration which may be seen as a possible test for the Ā-hypothesis involves verbal WH-movement. In A'ingae content questions, the WH-word is obligatorily fronted, and a second-position clitic must appear to its immediate right (Dąbkowski, 2022). As such, Dąbkowski (t.a.) analyzes the A'ingae P2 clitics as C-heads which trigger WH-movement.

In addition, A'ingae has a question verb *mikun* 'do what' which—like all other WH-words—must appear right before the P2 clitic. Since P2 clitics trigger WH-movement, they presumably have the uninterpretable [uWH] feature. Since the WH-verb *mikun* 'do what' can be inflected with situation-level morphology (such as the plural subject *-ʔfa* PLS and irrealis *-ya* IRR), it is a TP. After fronting, the P2 clitic forms one morphosyntactic (and phonological) word with the WH-verb, creating a configuration where T° is adjacent to a [uWH]-carrying morpheme (80). The span stretching from T° through the P2 clitic is underlined. Moved constituents and checked features are ~~stricken through~~.

(80) WH-VERB WITH A P2 CLITIC
[*míkun-ʔfa-ya-$\underline{\emptyset}$*]$_{TP}$ *$\underline{=tsû}$*$_{[\text{~~uWH~~}]}$ *~~míkun-ʔfa-ya~~*$_{[WH]}$

do what-PLS-IRR-T =3
"What will they do?" (2025-11-06(1)_jxm)

As seen in (80), the T-head is not realized as a glottal stop despite its adjacency to [~~uWH~~]. The interpretation of this fact depends on whether checked features, such [~~uWH~~], can be active at the point of allomorph selection. If one assumes that uninterpretable (checked) features can trigger allomorphy, the case observed in (80) counts as evidence against adopting the Ā-feature geometry of (79). If, on the other hand, checked features are inert for purposes of determining allomorphy, the structure in (80) is irrelevant to choosing between the two alternatives.

In either case, while the A'ingae data may not provide conclusive evidence with regard to the role of features such as [WH] or [REL], both analyses (T ⟷ *-ʔ* / _ δ as well as T ⟷ *-ʔ* / _ Ā) require discourse features to be organized in a geometrical hierarchy, with one higher-order feature dominating all the maximal ones.

More broadly, I observe that Ā-feature geometries, such as the one in (79), have been mostly motivated by data from languages where different Ā-carrying constituents pattern together in some way, e. g. with respect to long-distance WH-licensing (Aravind, 2018), morphological impoverishment (Baier, 2018), or Ā-movement. In A'ingae, it is not clear that the WH-feature has anything in common with the language's discourse markers. For example, the A'ingae WH-constituents undergo movement to Spec,CP (Dąbkowski, 2022, t.a.), but discourse-marked constituents do not (see §3). This raises the possibility that the Ā-feature geometry of (79) is not motivated for A'ingae.